

CHARACTERISTIC POLYNOMIALS OF RANDOM MATRICES AND $\zeta(1/2 + it)$

ABSTRACT. In this dissertation, we explore the connections between the statistical properties of the zeta function on the critical line and the characteristic polynomials of matrices in the CUE and other ensembles of Random Matrix Theory. We investigate the theoretical underpinnings and prove carefully most of the results in [1], which is the main resource and motivation for this thesis. We start by computing the generating functions of $\log Z(U, \theta)$ and we show each converges independently to a Gaussian distribution, allowing us to prove a central limit theorem which has a direct correspondent to a theorem of Selberg. Then we compute the leading-order asymptotics of each moment and derive exact expressions for each circular ensemble. It is conjectured that, in the CUE case, this quantity matches a factor from an important conjecture about the moments of zeta function high up the critical line. Moreover, Odlyzko's numerical data are used to establish further connections between the real and imaginary moments of the zeta function and the generating functions. We finally compute the value distribution of $|Z|$ at 0.



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0. NOTATION AND REMARKS

It is assumed that the reader is familiar with the notation and content of the relevant Part C courses (Analytic Number Theory, Random Matrix Theory), so the results taken from the lecture notes will be stated without citations. However, we include for completeness some specific notations and constants used in this thesis:

- $\int_{X^n}$ is shorthand for integrating n times $\int_X \cdots \int_X$.
- $\operatorname{meas}\{X\}$ is the measure of the set X in the Borel sense.
- $\mathbb{E}_G(f) = [f]_G$ for a compact Lie group G means $\int_G f d\mu$, where μ is the Haar measure.
- $U_\alpha(N)$ is the group of $N \times N$ unitary matrices with respect to the relevant ensemble (as defined later). Similarly, $\log Z(\alpha, U, \theta)$ and M_N^α, L_N^α denote the corresponding characteristic polynomial and moments.
- $\mathbf{1}_X$ is the indicator function of X .
- RHS is shorthand for "right hand side" and denotes the right term of an equality.
- $A \simeq 1.282$ is the Glaisher-Kinkelin constant.
- $\gamma \simeq 0.577 \dots$ is the Euler-Mascheroni constant.

Throughout this thesis, we have occasionally included very similar computations to the ones in [1], mostly for completeness, their relevance and/or because there were very few details worth adding, if any. The authors are given full credit for them.

1. INTRODUCTION

One of the most important conjectures in mathematics is the Riemann Hypothesis, which states that all non-trivial zeros of the zeta function

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_{p \text{ prime}} \left(1 - \frac{1}{p^s}\right)^{-1},$$

defined for $\operatorname{Re}(s) > 1$ and by analytic continuation to the rest of the complex plane, lie on the critical line $\operatorname{Re}(s) = 1/2$ (so they are of the form $1/2 + it_n$, for $t_n \in \mathbb{R}$).

The zeta function has many important properties. For instance, it has no non-trivial zeros in the region $\operatorname{Re}(s) > 1$. This follows from the Dirichlet series of $1/\zeta$, which converges absolutely in this range: $\frac{1}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\mu(n)}{n^s}$. Then we can evaluate the counting function of the non-trivial zeros

$$N(T) := \#\{n : 0 < \operatorname{Re}(t_n) \leq T\}$$

by integrating $\zeta'(s)/\zeta(s)$ along a thin rectangle containing them [2]. It is stated in [2] and it was first proven by Von Mangoldt [3] that

$$N(T) = \frac{T}{2\pi} \log \frac{T}{2\pi} - \frac{T}{2\pi} + \frac{7}{8} + O\left(\frac{1}{T}\right) + S(T), \text{ where } S(T) = O(\log T).$$

So $N(T) \sim \frac{T}{2\pi} \left(\log \frac{T}{2\pi} - 1\right) \sim \frac{T}{2\pi} \log \frac{T}{2\pi}$. Setting $w_n = \frac{t_n}{2\pi} \log \frac{|t_n|}{2\pi}$, we clearly have

$$\lim_{N \rightarrow \infty} \frac{1}{N(T)} \#\{n : 0 < n \leq T\} = 1 = \lim_{\substack{W \rightarrow \infty \\ W = \frac{T}{2\pi} \log \frac{T}{2\pi}}} \#\{n : 0 < w_n \leq W\},$$

since $\frac{x}{2\pi} \log \frac{x}{2\pi}$ is an increasing function for large x .

The first connection between random matrix theory and number theory has been made by Montgomery [4], who conjectured that, under the assumptions of the Riemann Hypothesis, the 2-point correlation function between the scaled zeros w_n is the same as the 2-kernel for the eigenvalues of large GUE random matrices (which are defined in 1.5). Specifically,

$$\lim_{W \rightarrow \infty} \frac{1}{W} \#\{w_m, w_n \in [0, W] : \alpha \leq w_n - w_m < \beta\} = \int_{\alpha}^{\beta} \left(\delta(x) + 1 - \frac{\sin^2(\pi x)}{\pi^2 x^2} \right) dx.$$

This is supported by a similar theorem, also proved by Montgomery in [4], which asserts that

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{m, n \leq N} f(w_n - w_m) = \int_{-\infty}^{\infty} f(x) \left(\delta(x) + 1 - \frac{\sin^2(\pi x)}{\pi^2 x^2} \right) dx,$$

for all test functions $f(x)$ for which the sum and integral converge and whose Fourier transforms

$$\hat{f}(\sigma) = \int_{-\infty}^{\infty} f(x) e^{2\pi i x \sigma} dx$$

have support in $(-1, 1)$. The generalized Montgomery conjecture states that the theorem holds for all test functions for which we have convergence, without any

conditions on the Fourier transform. In this case, the first equality follows by taking $f = \mathbb{1}_{(\alpha, \beta)}$.

The corresponding GUE correlations are the same as those of the phases of the eigenvalues, scaled so as to have unit mean spacing: $\phi_n = \theta_n \frac{N}{2\pi}$. This is proved in detail in Chapter 3 of [5]. Furthermore, the pair correlation results extend to all n -point correlations [6] and there is a lot of numerical evidence that supports the results in Odlyzko's monographs [Od3], [Od4] from [7] (we shall just cite [7] from now on when referring to them).

One of the aims of this paper is to present significant correlations inspired from the above results, inferred from numerical data and supported by theorems and conjectures, between the moments of the zeta function and the generating functions of the characteristic polynomials of unitary matrices. To this end, we investigate the value distribution of the characteristic polynomials

$$Z(U, \theta) = \det(I - Ue^{-i\theta}),$$

which can be thought of as an approximation to the zeta function for the RMT setting.

We start by recalling some useful definitions and theorems:

Definition 1.1. A matrix $U \in \mathcal{M}_n(\mathbb{C})$ is unitary if $UU^* = U^*U = I_n$, where U^* is the conjugate transpose of U . The eigenvalues have unit modulus, so they can be written as $e^{i\phi}$, for some $\phi \in [0, 2\pi)$, and the mean density of the eigenphases ϕ is $n/2\pi$.

Definition 1.2 (Haar measure). Let G be a locally compact Hausdorff topological group. Then a left-invariant Haar measure on G is a Borel measure μ satisfying the following conditions (the right-invariant one is defined similarly):

- $\mu(xE) = \mu(E)$, for every $x \in G$ and any measurable subset $E \subseteq G$;
- $\mu(U) > 0$, for every nonempty open subset $U \subseteq G$;
- $\mu(K) < \infty$, for every compact subset $K \subseteq G$.

Theorem 1.3. *Every compact Lie group has a unique left/right-invariant Haar measure (up to scalars). However, it may not automatically be bi-invariant. A proof can be found [here](#).*

Theorem 1.4. *Every compact Lie group is isomorphic to a matrix Lie group. This is a consequence of the [Peter-Weyl Theorem](#).*

Definition 1.5. The Gaussian Unitary Ensemble (GUE) is the class of Hermitian random matrices for which the entries are independent complex normal random variables. Specifically,

- M_{ij} are i.i.d. random variables for $i < j$, whose real and imaginary parts are independent Gaussians with mean 0 and variance 1/2;
- M_{ii} are i.i.d. Gaussian random variables, with mean 0 and variance 1.

We can define the Gaussian Orthogonal ensemble similarly, but this time we have symmetric random matrices and $N(0, 1)$ and $N(0, 2)$ distributions. It can be proved that the probability measures are indeed invariant under conjugation by unitary/orthogonal matrices.

Definition 1.6. The Circular Unitary Ensemble (CUE) is the group of $n \times n$ unitary matrices, equipped with the Haar measure. We can define the orthogonal (COE) and symplectic (CSE) ensembles similarly, for the corresponding matrix groups.

Definition 1.7. The cumulants c_n of a random variable X are obtained from the logarithm of the generating function (when it exists):

$$K(t) = \log \mathbb{E}[e^{tX}] = \sum_{n=1}^{\infty} c_n \frac{t^n}{n!},$$

from which it is straightforward to see that $c_n = K^{(n)}(0)$, $\forall n \geq 1$.

Definition 1.8. A class function on a matrix M is a function f which is symmetric in its arguments and depends only on the eigenvalues λ_i : $f(M) = f(\lambda_1, \dots, \lambda_n)$.

Theorem 1.9. *The average of f over the CUE is*

$$\begin{aligned} \mathbb{E}_{U(n)}[f(\lambda_1, \dots, \lambda_n)] &= [f(\lambda_1, \dots, \lambda_n)]_{U(n)} \\ &= \frac{1}{(2\pi)^n n!} \int_{[0, 2\pi]^n} f(\lambda_1, \dots, \lambda_n) \prod_{j < k} |e^{i\lambda_j} - e^{i\lambda_k}|^2 d\lambda_1 \dots d\lambda_n, \end{aligned}$$

which is a particular case of the Weyl Integration Formula.

As stated above, we explore the connection between the statistical properties of $\zeta(s)$ high up the critical line and the corresponding behaviour of $Z(U, \theta)$ averaged over the CUE. Specifically, we have the following limit theorem [8], which also has an informal proof [here](#):

Theorem 1.10 (Selberg). *For any rectangle $R \in \mathbb{R}^2$,*

$$\lim_{T \rightarrow \infty} \frac{1}{T} \text{meas} \left\{ t : T \leq t \leq 2T, \frac{\log \zeta(1/2 + it)}{\sqrt{1/2 \log \log T}} \in R \right\} = \frac{1}{2\pi} \iint_R e^{-(x^2 + y^2)/2} dx dy,$$

i.e. the distributions of the real and imaginary parts of $\frac{\log \zeta(1/2 + it)}{\sqrt{1/2 \log \log T}}$ tend to a Gaussian random variable with mean 0 and variance 1.

We will prove in the following sections a similar result [1]:

Theorem 1.11 (Keating, Snaith).

$$\frac{\log Z}{\sqrt{1/2 \log N}} \rightarrow X + iY,$$

where X and Y are i.i.d. $N(0, 1)$ (we shall see that the theorem holds irrespective of θ). Equivalently,

$$\lim_{N \rightarrow \infty} \text{meas} \left\{ M \in U(N) : \frac{\log Z(U, \theta)}{\sqrt{1/2 \log N}} \in R \right\} = \frac{1}{2\pi} \iint_R e^{-(x^2 + y^2)/2} dx dy,$$

where the real and imaginary parts of $\log Z$ converge independently in distribution.

For our purposes, the imaginary part of the logarithm is defined [9] by choosing the branch that satisfies $\log Z(U, \theta) = \sum_{n=1}^N \log(1 - e^{i(\theta_n - \theta)})$, as follows from the 'usual' factorisation of the characteristic polynomial, and $\text{Im} \log(1 - e^{i(\theta_n - \theta)}) \in (-\pi/2, \pi/2]$, for each $1 \leq n \leq N$.

We will also derive asymptotic formulae for the cumulants of the two distributions. This corresponds directly to 1.10 if we identify $\frac{N}{2\pi}$, the mean density of the eigenangles

of $Z(U, \theta)$, with the mean density of the Riemann zeros up the critical line, $\frac{1}{2\pi} \log \frac{T}{2\pi}$.

Number theorists have long been intrigued by the following assertion ([1]):

Conjecture 1.12. There exists a function $f(\lambda)$, defined via

$$\lim_{N \rightarrow \infty} \frac{1}{(\log N)^{\lambda^2}} \frac{1}{N} \int_0^N |\zeta(1/2 + it)|^{2\lambda} dt = f(\lambda) a(\lambda), \text{ where}$$

$$a(\lambda) = \prod_{p \text{ prime}} \left[(1 - 1/p)^{\lambda^2} \left(\sum_{n=0}^{\infty} \left(\frac{\Gamma(\lambda + n)}{n! \Gamma(\lambda)} \right)^2 p^{-n} \right) \right],$$

for all λ for which the moments are defined. It is an even bigger problem to determine the values of f , particularly for integers.

Remark 1.13. $f(0) = 1$.

Proof. For $\lambda = 0$, $a(0) = \lim_{\lambda \rightarrow 0} a(\lambda)$ and we get from 1.12 that $f(0)a(0) = 1$. Moreover, $\Gamma(\lambda + n) = \Gamma(\lambda) \{(\lambda + n - 1)(\lambda + n - 2) \dots \lambda\}$, so $\Gamma(\lambda + n)/(n! \Gamma(\lambda)) = \delta_{n0}$ in the limit $\lambda \rightarrow 0$ and we are left with $a(0) = 1$. So $f(0) = 1$. ■

If this turns out to be true, we could also prove the [Lindelöf Hypothesis](#), which states that, for any $\epsilon > 0$,

$$\zeta\left(\frac{1}{2} + it\right) = O(t^\epsilon), \text{ or, equivalently,}$$

$$\frac{1}{T} \int_0^T |\zeta(1/2 + it)|^{2k} dt = O(t^\epsilon),$$

for all positive integers k and all $\epsilon > 0$. This would bring us a step closer to the proof of the Riemann Hypothesis. Backlund has shown [10] that the Lindelöf Hypothesis is equivalent to the statement that the number of zeros with real part at least $1/2 + \epsilon$ and imaginary part between T and $T + 1$ is $o(\log T)$, as $T \rightarrow \infty$.

It is known that $f(1) = 1$ [11], $f(2) = 1/12$ [12] and conjectured that $f(3) = 42/9!$ [13], $f(4) = 24024/16!$ [14]. In fact, we expect to have

$$f(k) = \prod_{j=0}^{k-1} \frac{j!}{(j+k)!}, \quad \forall k \geq 1.$$

This guess is inspired by [1]

Theorem 1.14.

$$f_{\text{CUE}}(\lambda) = \lim_{N \rightarrow \infty} \frac{1}{N^{\lambda^2}} [Z(U, \theta)^{2\lambda}]_{U(N)} = \frac{G^2(1 + \lambda)}{G(1 + 2\lambda)},$$

where G is the Barnes G -function (defined later), and the authors claim that

Conjecture 1.15. $f(\lambda) = f_{\text{CUE}}(\lambda)$,

for all λ where the moments are defined, which is strongly supported by Odlyzko's numerical data [7]. These are in turn motivated by 1.12, by observing that replacing each p by p^k yields $a(\lambda) \rightarrow 1$ as $k \rightarrow \infty$.

This would imply that, in the conditions of Conjecture 1.12, the scaled moments of the zeta function high up the critical line split up into a factor which is specific

to the Riemann zeta function and the corresponding moment of the characteristic polynomial $Z(U, \theta)$ for large CUE matrices.

Finally, we prove two very important theorems in Section 2 [1]:

Theorem 1.16 (Keating, Snaith). *For $\text{Re}(s) > -1$,*

$$M_N(s) = [|Z(U, \theta)|^s]_{U(N)} = \prod_{j=1}^N \frac{\Gamma(j)\Gamma(j+s)}{(\Gamma(j+s/2))^2},$$

where $[|Z(U, \theta)|^s]_{U(N)} = [e^{s \log |Z(U, \theta)|}]_{U(N)} = [e^{s \text{Re} \log Z(U, \theta)}]_{U(N)}$ is the generating function of $\text{Re} \log Z$ and extends by analytic continuation to the entire complex plane. Similarly,

Theorem 1.17 (Keating, Snaith). *For $s \in \mathbb{C}$,*

$$L_N(s) = \left[\left(\frac{Z(U, \theta)}{Z^*(U, \theta)} \right)^{s/2} \right]_{U(N)} = \prod_{j=1}^N \frac{(\Gamma(j))^2}{\Gamma(j+s/2)\Gamma(j-s/2)},$$

with $Z(U, \theta) = |Z(U, \theta)|e^{i\phi}$, where $\phi = \phi(U, \theta) \in \arg Z(U, \theta)$. In this case, $L_N(s) = \left[(e^{i\phi}/e^{-i\phi})^{s/2} \right]_{U(N)} = [e^{is\phi}]_{U(N)} = [e^{is \text{Im} \log Z(U, \theta)}]_{U(N)}$, which is the generating function of $i \text{Im} \log Z$. The argument is defined in the same way as for $\log Z(U, \theta)$ above.

2. AVERAGES OF GENERATING FUNCTIONS OVER RANDOM MATRIX ENSEMBLES

The most important result of this section (and of great importance to RMT generally) is the [Weyl Integration Formula](#), from which we get the expressions for the probability densities of various matrix Lie groups (for instance, see [9] and [15]). To prove 1.16 and 1.17, we state a very useful result [16]:

Theorem 2.1 (Selberg's Formula).

$$\begin{aligned} J(a, b, \alpha, \beta, \gamma, N) &= \int_{\mathbb{R}^N} |\Delta(x)|^{2\gamma} \prod_{j=1}^N (a + ix_j)^{-\alpha} (b - ix_j)^{-\beta} dx_1 \dots dx_N \\ &= \frac{(2\pi)^N}{(a+b)^{(\alpha+\beta)N-\gamma N(N-1)-N}} \prod_{j=0}^{N-1} \frac{\Gamma(1+\gamma+j\gamma)\Gamma[\alpha+\beta-(N+j-1)\gamma-1]}{\Gamma(1+\gamma)\Gamma(\alpha-j\gamma)\Gamma(\beta-j\gamma)}, \end{aligned}$$

where $a, b, \alpha, \beta \in \mathbb{C}$; $\operatorname{Re}(a), \operatorname{Re}(b), \operatorname{Re}(\alpha), \operatorname{Re}(\beta) > 0$; $\operatorname{Re}(a+b) > 1$; $-\frac{1}{N} < \operatorname{Re}(\gamma) < \min\left(\frac{\operatorname{Re}(a)}{N-1}, \frac{\operatorname{Re}(b)}{N-1}, \frac{\operatorname{Re}(\alpha+\beta-1)}{2(N-1)}\right)$ and $\Delta(x)$ is the Vandermonde determinant given by the x_i , for $1 \leq i \leq N$.

Proof of 1.16. Let us now look at $Z(U, \theta)$ again. If U has eigenvalues $e^{i\theta_n}$, then we can write it as $Z(U, \theta) = \prod_{n=1}^N (1 - e^{i(\theta_n - \theta)})$. The joint probability density of the θ_n is $[(2\pi)^N N!] \prod_{j < m} |e^{i\theta_j} - e^{i\theta_m}|^2$, from the Weyl Integration Formula. This gives

$$[|Z|^s]_{U(N)} = \frac{1}{(2\pi)^N N!} \int_{[0, 2\pi]^N} \prod_{1 \leq j < k \leq N} |e^{i\theta_j} - e^{i\theta_k}|^2 \left| \prod_{n=1}^N (1 - e^{i(\theta_n - \theta)}) \right|^s d\theta_1 \dots d\theta_N.$$

But

$$\begin{aligned} |e^{i\theta_j} - e^{i\theta_k}|^2 &= |1 - e^{i(\theta_j - \theta_k)}|^2 = |2 \sin^2(\theta_j - \theta_k)/2 + 2i \sin(\theta_j - \theta_k)/2 \cos(\theta_j - \theta_k)/2| \\ &= 4 \sin^2(\theta_j - \theta_k)/2, \text{ and similarly } |1 - e^{i(\theta_j - \theta_k)}| = 2 |\sin(\theta_n - \theta)/2|. \end{aligned}$$

We get a $4^{N(N-1)/2}$ contribution from the first product and 2^{sN} from the second, so the first equality becomes

$$\begin{aligned} [|Z|^s]_{U(N)} &= [e^{s \operatorname{Re} \log Z(U, \theta)}]_{U(N)} = \\ &= \frac{2^{N(N-1)+sN}}{N!(2\pi)^N} \int_{[0, 2\pi]^N} \prod_{1 \leq j < k \leq N} |\sin(\theta_j - \theta_k)/2|^2 \prod_{n=1}^N |\sin(\theta_n - \theta)/2|^s d\theta_1 \dots d\theta_N. \end{aligned}$$

This integral is independent of θ , so we can set $\theta = 0$. Substituting $\psi_i = \theta_i/2$ and using the identity $\cot(\psi_j) - \cot(\psi_k) = \frac{\sin(\psi_k - \psi_j)}{\sin \psi_k \sin \psi_j}$, each $\sin \psi_i$ will be counted $(N-1)$ times when making the identification in the first product, so

$$\begin{aligned} [|Z|^s]_{U(N)} &= \\ &= \frac{2^{N^2+sN}}{N!(2\pi)^N} \int_{[0, \pi]^N} \prod_{1 \leq j < k \leq N} |\cot \psi_j - \cot \psi_k|^2 \prod_{n=1}^N (\sin^2 \psi_n)^{N-1} \prod_{i=1}^N |\sin \psi_i|^s d\psi_1 \dots d\psi_N \end{aligned}$$

$$\begin{aligned}
&= \frac{2^{N^2+sN}}{N!(2\pi)^N} \int_{\mathbb{R}^N} \prod_{1 \leq j < k \leq N} |x_j - x_k|^2 \prod_{n=1}^N (1 + x_n^2)^{-N-s/2} dx_1 \dots dx_N \\
&= \frac{2^{N^2+sN}}{N!(2\pi)^N} \int_{\mathbb{R}^N} |\Delta(x)|^2 \prod_{n=1}^N [(1 - ix_n)(1 + ix_n)]^{-N-s/2} dx_1 \dots dx_N,
\end{aligned}$$

upon making the change of variables $x_i = \cot \psi_i$. But this is a Selberg integral with $a = b = \gamma = 1, \alpha = \beta = N + s/2, N = N$ that fulfills the conditions and hence

$$\begin{aligned}
M_N(s) &= \frac{2^{N^2+sN}}{N!(2\pi)^N} J(1, 1, N + s/2, N + s/2, 1, N) = \prod_{j=1}^N \frac{\Gamma(j)\Gamma(s+j)}{(\Gamma(j+s/2))^2} \\
&= \frac{2^{N^2+sN}}{N!(2\pi)^N} \frac{(2\pi)^N}{2^{(2N+s)N-N(N-1)-N}} \prod_{k=0}^{N-1} \frac{\Gamma(2+k)\Gamma(2N+s-(N+k-1)-1)}{\Gamma(2)(\Gamma(N+s/2-k))^2} \\
&= \frac{1}{N!} \prod_{t=2}^{N+1} \Gamma(t) \prod_{j=1}^N \frac{\Gamma(s+j)}{(\Gamma(j+s/2))^2} = \prod_{j=1}^N \frac{\Gamma(j)\Gamma(s+j)}{(\Gamma(j+s/2))^2},
\end{aligned}$$

where we have used $\Gamma(n) = (n-1)!, \forall n \geq 1$, and the substitutions $t = k+2, j = N-k$, provided $\text{Re}(s) > -1$. ■

Proof of 1.17. Let us now consider the other generating function. From (16), (18) and by expanding the exponential in Z we have

$$\begin{aligned}
[e^{is\text{Im} \log Z}]_{U(N)} &= \left[\exp \left(i \sum_{n=1}^N \text{Im} \log (1 - e^{i(\theta_n - \theta)}) \right) \right]_{U(N)} \\
&= \left[\exp \left(-i \sum_{n=1}^N \text{Im} \sum_{k=1}^{\infty} \frac{e^{i(\theta_n - \theta)k}}{k} \right) \right]_{U(N)} \\
&= \left[\exp \left(-i \sum_{n=1}^N \sum_{k=1}^{\infty} \frac{\sin [(\theta_n - \theta)k]}{k} \right) \right]_{U(N)},
\end{aligned}$$

by using the expansion $\log(1-z) = -\sum_{k=1}^{\infty} \frac{z^k}{k}$, valid for $|z| < 1$. Combining (2.1) with the joint probability density of the N eigenangles θ_i , the identity $\sum_{k=1}^{\infty} \frac{\sin kx}{k} = \frac{\pi-x}{2}$, for $0 < x < 2\pi$ (proved in A.1 and setting $\theta = 0$ again (as the average over $U(N)$ has spherical symmetry), we get

$$\begin{aligned}
L_N(s) &= \left[\exp \left(-is \sum_{n=1}^N \sum_{k=1}^{\infty} \frac{\sin [(\theta_n - \theta)k]}{k} \right) \right]_{U(N)} \\
&= \frac{2^{N(N-1)}}{N!(2\pi)^N} \int_{[0,2\pi]^N} \prod_{1 \leq j < k \leq N} \left| \sin \frac{\theta_j - \theta_k}{2} \right|^2 \prod_{n=1}^N \exp \left\{ \left(-\frac{is}{2} (\pi - \theta_n) \right) \right\} d\theta_1 \dots d\theta_N.
\end{aligned}$$

Now substitute $\psi_k = \theta_k/2 - \pi/2$ and use the identity $\sin(\psi_j - \psi_k) = (\tan \psi_j - \tan \psi_k) \cos \psi_j \cos \psi_k$, which follows immediately upon expanding the sine. We get a further 2^N contribution from the $d\theta_i$. After that, take $x_k = \tan \psi_k$, for $1 \leq k \leq N$.

Then

$$\begin{aligned}
L_N(s) &= \frac{2^{N^2}}{N!(2\pi)^N} \int_{[-\pi/2, \pi/2]^N} \prod_{1 \leq j < k \leq N} |\tan \psi_j - \tan \psi_k|^2 \\
&\quad \prod_{n=1}^N (\cos^2 \psi_n)^{N-1} \prod_{k=1}^N (\cos \psi_k + i \sin \psi_k)^s d\psi_1 \dots d\psi_N = \\
&= \frac{2^{N^2}}{N!(2\pi)^N} \int_{\mathbb{R}^N} \prod_{1 \leq j < k \leq N} |x_j - x_m|^2 \prod_{n=1}^N \left(\frac{1}{1+x_n^2} \right)^N \prod_{k=1}^N \left(\frac{1+ix_k}{\sqrt{1+x_k^2}} \right)^s \\
&\quad dx_1 \dots dx_N = \\
&= \frac{2^{N^2}}{N!(2\pi)^N} \int_{\mathbb{R}^N} \prod_{1 \leq j < k \leq N} |x_j - x_m|^2 \prod_{n=1}^N \left(\frac{1}{1+x_n^2} \right)^N \prod_{k=1}^N \left(\sqrt{\frac{1+ix_k}{1-ix_k}} \right)^s \\
&\quad dx_1 \dots dx_N = \\
&= \frac{2^{N^2}}{N!(2\pi)^N} \int_{\mathbb{R}^N} \prod_{1 \leq j < k \leq N} |\Delta(x)|^2 \prod_{n=1}^N (1+ix_n)^{-N+s/2} (1-ix_n)^{-N-s/2} dx_1 \dots dx_N.
\end{aligned}$$

Here we have used basic trigonometric transformations involving the tangent and changed the limits of the integral. We can safely do so, as the integrals exist and they are of the Selberg type. Indeed, we can take $a = b = 1$, $\alpha = N - s/2$, $\beta = N + s/2$ and $\gamma = 1$ in 2.1. The last condition is satisfied when $|s| < 2$. We end up with

$$\begin{aligned}
L_N(s) &= \frac{2^{N^2}}{N!(2\pi)^N} J(1, 1, N - s/2, N + s/2, 1) \\
&= \frac{(2\pi)^N}{2^{2N^2 - N(N-1) - N}} \frac{2^{N^2}}{N!(2\pi)^N} \prod_{k=0}^{N-1} \frac{\Gamma(2+k)\Gamma(2N - (N+k-1) - 1)}{\Gamma(2)\Gamma(N - s/2 - k)\Gamma(N + s/2 - k)} \\
&= \frac{1}{N!} \prod_{t=2}^{N+1} \Gamma(t) \prod_{j=1}^N \frac{\Gamma(j)}{\Gamma(j - s/2)\Gamma(j + s/2)} = \prod_{j=1}^N \frac{(\Gamma(j))^2}{\Gamma(j - s/2)\Gamma(j + s/2)},
\end{aligned}$$

where we have the same substitutions and conditions as in the previous proof. ■

3. VALUE DISTRIBUTIONS OF $\log Z$ AND PROOF OF THE LIMIT THEOREM FOR $Z(U, \theta)$

Our next endeavour is to prove Theorem 1.11, which we state again:

$$\lim_{N \rightarrow \infty} \text{meas} \left\{ M \in U(N) : \frac{\log Z(U, \theta)}{\sqrt{1/2 \log N}} \in R \right\} = \frac{1}{2\pi} \iint_R e^{-(x^2+y^2)/2} dx dy.$$

It is worth commenting about the differences between 1.11 and the classical central limit theorem for i.i.d. random variables. As we will see from 3.1, it follows that the first cumulant (which is equal to the mean) vanishes, but $\log Z$ does not scale like $\sqrt{N}$. This is because the eigenangles of CUE matrices, as the eigenvalues of GUE matrices, tend to the distribution given by the sine kernel determinantal point process (for more details, see section 1.3 of [17]). This means that the eigenangles are more evenly distributed (there is more repulsion) compared to a Poisson point process (which is the expected behaviour), so we need an asymptotically smaller scale to get the limiting Gaussian distribution.

The most direct approach is to try to find two distributions in the CUE which converge independently to the standard normal distribution. As we will soon find out, the scaled real and imaginary parts of $\log Z(U, \theta)$ fulfill these conditions.

3.1. Value distribution of $\text{Re } \log Z$. Let us look at $M_N(s)$ again. By Taylor expanding about $s = 0$,

$$M_N(s) = [e^{s \log Z}]_{U(N)} = \sum_{k=0}^{\infty} \frac{s^k [(\log Z)^k]_{U(N)}}{k!},$$

so the k^{th} moment is given by $\left. \frac{d^k}{ds^k} \right|_{s=0} M_N(s)$, $\forall k \geq 1$, and the cumulants by the expansion of $\log M_N(s)$,

$$\log M_N(s) = \sum_{k=1}^{\infty} \frac{c_k(N)}{k!} s^k.$$

It is useful to look at the probability density

$$\rho_N(x) = [\delta(\log |Z| - x)]_{U(N)},$$

from which changing variables will lead us to the normal distribution. The integral representation of the delta function gives

$$\begin{aligned} \rho_N(x) &= \frac{1}{2\pi} \left[\int_{-\infty}^{\infty} e^{iy(\log |Z| - x)} dy \right]_{U(N)} = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{iyx} [|Z|^{-iy}]_{U(N)} dy \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-iyx} M_N(iy) dy. \end{aligned}$$

We now prove an identity which makes use of the **polygamma function** $\psi^{(n)}(z) = \frac{d^{n+1}}{dz^{n+1}} \log \Gamma(z)$, that has integral representation $\psi^{(n)}(z) = (-1)^{n+1} \int_0^{\infty} \frac{t^n e^{-zt}}{1-e^{-t}} dt$, for $n > 0$ and $\text{Re}(z) > 0$.

Lemma 3.1. *The n^{th} cumulant satisfies*

$$c_n(N) = \frac{2^{n-1} - 1}{2^{n-1}} \sum_{k=1}^N \psi^{(n-1)}(k), \quad \forall n \geq 1.$$

Proof. Recall that $M_N(s) = \prod_{k=1}^N \frac{\Gamma(k)\Gamma(s+k)}{(\Gamma(s+k/2))^2}$ and definition 1.7. Moreover,

$$\log M_N(s) = \sum_{k=1}^N [\log \Gamma(k) + \log \Gamma(k+s) - 2 \log \Gamma(k+s/2)]$$

and only the logarithms containing s contribute to the expression for the cumulants. Now clearly

$$\left. \frac{d}{ds} \log \Gamma(k+s\alpha) \right|_{s=0} = \alpha \frac{d}{d(k+s\alpha)} \log \Gamma(k+s\alpha) \Big|_{s=0} = \alpha \psi^{(0)}(k),$$

$\forall 1 \leq k \leq N$ and $\alpha \in \mathbb{R}^*$. So it follows easily by induction that

$$\left. \frac{d^n}{ds^n} \log \Gamma(k+s\alpha) \right|_{s=0} = \alpha^n \frac{d^n}{d(k+s\alpha)^n} \log \Gamma(k+s\alpha) \Big|_{s=0} = \alpha^n \psi^{(n-1)}(k), \quad \forall n \geq 1.$$

Setting $\alpha = 1$ and $\alpha = 1/2$ in the expansion of $\log M_N(s)$, we are left with

$$c_n(N) = \sum_{k=1}^N \left(1 - 2 \times \frac{1}{2^n}\right) \psi^{(n-1)}(k) = \frac{2^{n-1} - 1}{2^{n-1}} \sum_{k=1}^N \psi^{(n-1)}(k), \quad \text{as required.}$$

■

We now use the integral representation of the polygamma function stated above to derive asymptotics for $c_2(N)$ and $c_n(N)$ (the former will prove very useful for obtaining the relevant distributions).

Proposition 3.2. *For $n \geq 2$ we have*

$$c_n(N) = \frac{2^{n-1} - 1}{2^{n-1}} (-1)^n \int_0^\infty \frac{e^{-t}}{1 - e^{-t}} [(n-1)t^{n-2} - (n-1)t^{n-2}e^{-Nt} + Nt^{n-1}e^{-Nt}] dt.$$

Proof. Use 3.1:

$$\begin{aligned} c_n(N) &= \frac{2^{n-1} - 1}{2^{n-1}} \sum_{k=1}^N (-1)^n \int_0^\infty \frac{t^{n-1} e^{-kt}}{1 - e^{-t}} dt \\ &= \frac{2^{n-1} - 1}{2^{n-1}} (-1)^n \int_0^\infty \frac{t^{n-1} e^{-t} (1 - e^{-Nt})}{(1 - e^{-t})^2} dt \\ &= \frac{2^{n-1} - 1}{2^{n-1}} (-1)^n \int_0^\infty \frac{[t^{n-1} (1 - e^{-Nt})]'}{1 - e^{-t}} - \left(\frac{t^{n-1} (1 - e^{-Nt})}{1 - e^{-t}} \right)' dt \\ &= \frac{2^{n-1} - 1}{2^{n-1}} (-1)^n \left\{ [t^{n-1} (1 - e^{-Nt})]_{t=0}^\infty - \left[\frac{t^{n-1} (1 - e^{-Nt})}{1 - e^{-t}} \right]_{t=0}^\infty \right\} + \\ &\quad + \frac{2^{n-1} - 1}{2^{n-1}} (-1)^n \int_0^\infty \frac{e^{-t}}{1 - e^{-t}} [(n-1)t^{n-2} - (n-1)t^{n-2}e^{-Nt} + Nt^{n-1}e^{-Nt}] dt. \end{aligned}$$

Here we have taken the integral over the sum and then integrated by parts (we can do this, since all the integrals converge). Both $t^{n-1} (1 - e^{-Nt}) \rightarrow 0$ and $\frac{t^{n-1} (1 - e^{-Nt})}{1 - e^{-t}} \rightarrow 0$ when $t \rightarrow 0$. Furthermore,

$$\lim_{t \rightarrow \infty} \frac{t^{n-1} (1 - e^{-Nt})}{1 - e^{-t}} - t^{n-1} (1 - e^{-Nt}) = \lim_{t \rightarrow \infty} \frac{t^{n-1} (1 - e^{-Nt}) e^{-t}}{1 - e^{-t}} = 0,$$

since $\lim_{t \rightarrow \infty} t^{n-1} e^{-t} = 0$ and the other factors tend to 1. So the curly brackets vanish and the conclusion follows. $\blacksquare$

The integral in 3.2 helps us derive an asymptotic expression for the n^{th} cumulant:

Proposition 3.3. *For $n \geq 3$,*

$$c_n(N) = \frac{2^{n-1} - 1}{2^{n-1}} (-1)^n \left(\Gamma(n) \zeta(n-1) - \frac{(n-3)!}{N^{n-2}} \right) + O(N^{1-n}).$$

Proof. Rewrite the cumulant 3.2 as

$$c_n(N) = \frac{2^{n-1} - 1}{2^{n-1}} (-1)^n \int_0^\infty \left((n-1) \frac{t^{n-2}}{e^t - 1} + (Nt^{n-1} - (n-1)t^{n-2}) \frac{e^{-Nt}}{e^t - 1} \right) dt,$$

and, from the integral representation $\Gamma(s)\zeta(s) = \int_0^\infty \frac{x^{s-1}}{e^x - 1} dx$ (which follows by expanding $\frac{1}{e^x - 1}$ and the definition of Γ) we have

$$c_n(N) = \frac{2^{n-1} - 1}{2^{n-1}} (-1)^n \left(\Gamma(n) \zeta(n-1) + \int_0^\infty (Nt^{n-1} - (n-1)t^{n-2}) \frac{e^{-Nt}}{e^t - 1} dt \right).$$

Now substitute $u = tN$:

$$\begin{aligned} c_n(N) &= \frac{2^{n-1} - 1}{2^{n-1}} (-1)^n \Gamma(n) \zeta(n-1) + \\ &\quad + \left(\frac{1}{N^{n-1}} \int_0^\infty (u^{n-1} - (n-1)u^{n-2}) e^{-u} \frac{1}{e^{u/N} - 1} du \right) \end{aligned}$$

Recall the definition of Bernoulli numbers in A.2: we can expand $\frac{1}{e^x - 1} = \sum_{n=0}^\infty \frac{B_n}{n!} x^{n-1}$, for $x \neq 0$. Taking $x = u/N$ and plugging this back into the integral,

$$\begin{aligned} I &= \frac{1}{N^{n-1}} \int_0^\infty (u^{n-1} - (n-1)u^{n-2}) e^{-u} \frac{1}{e^{u/N} - 1} du \\ &= \frac{1}{N^{n-1}} \int_0^\infty (u^{n-1} - (n-1)u^{n-2}) e^{-u} \sum_{k=0}^\infty \frac{B_k}{k!} \frac{u^{k-1}}{N^{k-1}} du \\ &= \frac{1}{N^{n-1}} \int_0^\infty (u^{n-1} - (n-1)u^{n-2}) e^{-u} \frac{N}{u} du + O(N^{1-n}), \end{aligned}$$

for large N , since all the integrals converge and $B_0 = 1$. In fact, it is easy to prove by integration by parts that $\int_0^\infty x^n e^{-x} dx = n!$, $\forall n \geq 1$. So we are left with

$$\begin{aligned} I &= \frac{1}{N^{n-2}} \int_0^\infty (u^{n-2} - (n-1)u^{n-3}) e^{-u} du \\ &= \frac{1}{N^{n-2}} [(n-2)! - (n-1)(n-3)!] = -\frac{(n-3)!}{N^{n-2}}, \end{aligned}$$

which concludes the proof. $\blacksquare$

Remark 3.4. Other expansions are possible. If we plug $\frac{1}{x-1} = \sum_{k=1}^{\infty} x^{-k}$, for $x > 1$ and $\int_0^{\infty} x^n e^{-\alpha x} = \frac{n!}{\alpha^{n+1}}$, for $\alpha > 0$ into the substituted integral, we get

$$\begin{aligned}
I &= \frac{1}{N^{n-1}} \int_0^{\infty} (u^{n-1} - (n-1)u^{n-2}) \sum_{k=1}^{\infty} e^{-u(k/N+1)} du \\
&= \frac{1}{N^{n-1}} \left[\sum_{k=1}^{\infty} \frac{(n-1)!}{(k/N+1)^n} - \sum_{k=1}^{\infty} \frac{(n-1)!}{(k/N+1)^{n-1}} \right] \\
&= \frac{1}{N^{n-1}} \sum_{k=1}^{\infty} \frac{(n-1)!N^n}{(k+N)^n} \left(-\frac{k}{N} \right) = -(n-1)! \sum_{k=1}^{\infty} \frac{k}{(k+N)^n} \\
&= -(n-1)! \left(N\zeta(n) - \zeta(n-1) + \sum_{k=1}^N \frac{1}{k^{n-1}} - \sum_{k=1}^N \frac{1}{k^n} \right),
\end{aligned}$$

from which we can derive other asymptotic estimates.

Consider $c_2(N)$ now. 3.2 gives the simpler expression

$$c_2(N) = \frac{1}{2} \int_0^{\infty} \left(\frac{1-e^{-Nt}}{1-e^{-t}} e^{-t} + Nt \frac{e^{-(N+1)t}}{1-e^{-t}} \right) dt,$$

from which we can re-expand the fractions and get

$$\begin{aligned}
c_2(N) &= \frac{1}{2} \int_0^{\infty} \left((1 + e^{-t} + \dots + e^{-(N-1)t}) e^{-t} - Nt \frac{1-e^{-(N+1)t}}{1-e^{-t}} + \frac{Nt}{1-e^{-t}} \right) dt \\
&= \frac{1}{2} \int_0^{\infty} \left(\sum_{k=1}^N e^{-kt} - Nt \sum_{k=0}^N e^{-kt} + Nt \sum_{k=0}^{\infty} e^{-kt} \right) dt \\
&= \frac{1}{2} \sum_{k=1}^N \int_0^{\infty} e^{-kt} dt + \frac{N}{2} \sum_{k=N+1}^{\infty} \int_0^{\infty} t e^{-kt} dt \\
&= \frac{1}{2} \sum_{k=1}^N \frac{1}{k} + \frac{N}{2} \sum_{k=N+1}^{\infty} \frac{1}{k^2},
\end{aligned}$$

where we have used the geometric series expansion in the second line and an integral result from 3.3 in the fourth. Everything converges, so we can exchange sums and integrals freely. A.3 and A.4 give the asymptotic expression for the first term:

$$\sum_{k=1}^n \frac{1}{k} = \gamma + \log n + \frac{1}{2n} - \sum_{k=2}^{\infty} \frac{A_k}{n(n+1) \dots (n+k-1)}.$$

We can apply A.2 again to $\sum_{k=N+1}^{\infty} \frac{1}{k^2}$ and compute any number of terms in the expansion. For instance,

$$c_2(N) = \frac{1}{2} \log N + \frac{1}{2}(\gamma + 1) + \frac{1}{24N^2} - \frac{1}{80N^4} + O\left(\frac{1}{N^6}\right).$$

As proved in A.5, $\frac{N}{2} \sum_{n=N+1}^{\infty} \frac{1}{n^2} \rightarrow \frac{1}{2}$ as $N \rightarrow \infty$, so it becomes negligible for large N and hence we derive a very important property,

$$c_2(N) \sim \frac{1}{2} \log N.$$

It is clear from 3.3 that $c_n(N) \sim \frac{2^{n-1}-1}{2^{n-1}}(-1)^n \Gamma(n) \zeta(n-1)$, for $n \geq 3$ and large N . It follows that $\frac{c_n(N)}{c_2(N)^{n/2}} \rightarrow 0$ as $N \rightarrow \infty$, $\forall n \geq 1$.

Consider the value distribution $\rho_n(x)$ again. Clearly $\int_{-\infty}^{\infty} \rho_n(y) dy = 1$. Letting $y = \sqrt{c_2(N)}x$ and changing variables gives $\int_{-\infty}^{\infty} \rho_n(\sqrt{c_2(N)}x) \sqrt{c_2(N)} dx = 1$. So it is enough to show that $\tilde{\rho}_n(x) := \sqrt{c_2(N)} \rho_n(\sqrt{c_2(N)}x)$ converges to the standard normal distribution as $N \rightarrow \infty$. Our next result is extremely useful:

Proposition 3.5.

$$\tilde{\rho}_N(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \exp \left(-iyx - \frac{y^2}{2} + \sum_{k=3}^{\infty} \frac{(iy)^k c_k(N)}{k! (c_2(N))^{k/2}} \right) dy.$$

Proof. By the explicit formula for $\rho_N(x)$ and applying the same substitution as before,

$$\begin{aligned} \tilde{\rho}_N(x) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-iyx} M_N \left(\frac{iy}{\sqrt{c_2(N)}} \right) dy \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-iyx} \exp \left(\log M_N \left(\frac{iy}{\sqrt{c_2(N)}} \right) \right) dy \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-iyx} \exp \left(\sum_{k=1}^{\infty} \frac{c_k(N) \left(iy / \sqrt{c_2(N)} \right)^k}{k!} \right) dy \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \exp \left(-iyx - \frac{y^2}{2} + \sum_{k=3}^{\infty} \frac{(iy)^k c_k(N)}{k! (c_2(N))^{k/2}} \right) dy, \end{aligned}$$

where we have used the definition of the cumulants in the third line. ■

Theorem 3.6.

$$\lim_{N \rightarrow \infty} \tilde{\rho}_N(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

Proof. As stated above, $c_n(N)/c_2(N)^{n/2} \rightarrow 0$ as $N \rightarrow \infty$ for every $n \geq 1$, so only the first two terms remain when taking the limit, i.e.

$$\begin{aligned} \lim_{N \rightarrow \infty} \tilde{\rho}_N(x) &= \int_{-\infty}^{\infty} \exp \left(-iyx - \frac{y^2}{2} \right) dy = \frac{1}{2\pi} \int_{-\infty}^{\infty} \exp \left(-\frac{x^2}{2} - \left(\frac{y+ix}{\sqrt{2}} \right)^2 \right) dy \\ &= \frac{i}{2\pi} \exp \left(\frac{-x^2}{2} \right) \int_{-i\infty}^{i\infty} \exp \left(\frac{t^2}{2} \right) dt \\ &= \frac{1}{2\pi} \exp \left(\frac{-x^2}{2} \right) \int_{\infty}^{-\infty} \exp \left(\frac{-u^2}{2} \right) du = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \end{aligned}$$

as required. Here we have substituted $t = x - iy$ and $u = it$. We can change the limits as in the real case (provided the integrals exist, which is true by the above calculations), because the imaginary axis can be thought as a copy of $\mathbb{R}$. ■

We can extract more information from the explicit formula in 3.5 by writing the exponential as a power series:

$$\begin{aligned}\tilde{\rho}_N(x) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \exp\left(-iyx - \frac{y^2}{2}\right) \left[\sum_{p=0}^{\infty} \frac{1}{p!} \left(\sum_{k=3}^{\infty} \frac{(iy)^k c_k(N)}{k! (c_2(N))^{k/2}} \right)^p \right] dy \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \exp\left(-iyx - \frac{y^2}{2}\right) dy + \frac{1}{2\pi} \int_{-\infty}^{\infty} \exp\left(-iyx - \frac{y^2}{2}\right) \left\{ \left(\frac{c_3(N)(iy)^3}{3! c_2(N)^{3/2}} + \right. \right. \\ &\quad \left. \left. + \frac{c_4(N)(iy)^4}{4! c_2(N)^2} + \dots \right) + \frac{1}{2!} \left(\frac{c_3(N)(iy)^3}{3! c_2(N)^{3/2}} + \frac{c_4(N)(iy)^4}{4! c_2(N)^2} + \dots \right)^2 + \dots \right\} dy\end{aligned}$$

We can now expand every term in the curly brackets and group those who have the same power of the second cumulant in the denominator (ignoring the factorial coefficients). Let $C_k(N)$ be the coefficient corresponding to $c_2(N)^{k/2}$. For example, $C_3(N) = \frac{1}{3!}$, since we only have only one fraction containing $c_2^{3/2}$. So

$$\tilde{\rho}_N(x) = \frac{1}{\sqrt{2\pi}} \exp\left(\frac{x^2}{2}\right) + \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-iyx - y^2/2} \left(\frac{C_3(N)(iy)^3}{c_2(N)^{3/2}} + \frac{C_4(N)(iy)^4}{c_2(N)^2} + \dots \right) dy.$$

Integrating term by term boils down to evaluating integrals of the form

$$I_n = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-iyx - y^2/2} (iy)^n dy.$$

These can be computed almost as in the proof of 3.6. Specifically,

$$\begin{aligned}I_n &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left(-\left(\frac{y+ix}{\sqrt{2}}\right)^2\right) (iy)^n dy \\ &= \frac{i}{2\pi} \int_{-i\infty}^{i\infty} e^{t^2/2} (t+x)^n dt \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-u^2/2} \sum_{p=0}^n x^p i^{n-p} \binom{n}{p} u^{n-p} du \\ &= \sum_{p=0}^n \binom{n}{p} x^p i^{n-p} \int_{-\infty}^{\infty} e^{-u^2/2} u^{n-p} du.\end{aligned}$$

Now we have to consider the parity of $n - p$. Since $e^{x^2/2} x^{n-p}$ is an odd function for $n - p$ odd, the integral will vanish in this case. For n even, let $J_k = \int_{-\infty}^{\infty} e^{-x^2/2} x^{2k} dx$, for $k \geq 0$. Integrating by parts,

$$\begin{aligned}J_k &= - \int_{-\infty}^{\infty} \left(e^{-x^2/2} \right)' x^{2k-1} dx \\ &= - \left[e^{-x^2/2} x^{2k-1} \right]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} (2k-1) e^{-x^2/2} x^{2k-2} dx \\ &= (2k-1) J_{k-1}.\end{aligned}$$

The square brackets vanish because $\lim_{n \rightarrow \pm\infty} \frac{x^{2k-1}}{e^{x^2/2}} = 0$, which can be computed by applying L'Hôpital's Rule repeatedly. It is then straightforward to show by induction that

$$J_k = \{(2k-1)(2k-3)\dots 1\} J_0 = (2k-1)!!\sqrt{2\pi}, \forall k \geq 1.$$

Plugging this back into the main expression gives

$$\tilde{\rho}_N(x) = \frac{1}{\sqrt{2\pi}} \exp\left(\frac{x^2}{2}\right) + \frac{1}{\sqrt{2\pi}} \sum_{n=3}^{\infty} \frac{A_n(N)}{c_2(N)^{n/2}} e^{-x^2/2} \sum_{p=0}^{\infty} \binom{n}{p} x^p i^{n-p} (n-p-1)!!$$

for n even and 0 for n odd.

We can use the previous formulae to obtain the last important result of this section:

Proposition 3.7. *For every $n \geq 1$,*

$$\begin{aligned} [(\log |Z|)^n]_{U(N)} &= \frac{d^n}{ds^n} M_N(s) \Big|_{s=0} \\ &= \begin{cases} (2k-1)!! [(\log |Z|)^2]_{U(N)}^k & \text{if } n = 2k \\ O((\log N)^{k-1}) & \text{if } n = 2k+1. \end{cases} \end{aligned}$$

Proof. Rewrite $M_N(s)$ in two different ways. We have

$$M_N(s) = \sum_{n=0}^{\infty} [(\log |Z|)^n] \frac{s^n}{n!} \text{ and also}$$

$$M_N(s) = e^{\log M_N(s)} = \exp\left(\sum_{m=1}^{\infty} c_m(N) \frac{s^m}{m!}\right) = \sum_{k=0}^{\infty} \frac{1}{k!} \left(\sum_{m=2}^{\infty} c_m(N) \frac{s^m}{m!}\right)^k,$$

since the first cumulant vanishes.

We will prove later in 3.13 that all cumulants, except for the second one, are $O(1)$. This means that we can write $M_N(s)$ as

$$\begin{aligned} M_N(s) &= \frac{1}{k!} \left\{ \left(c_2(N) \frac{s^2}{2}\right)^k + \left(c_2(N) \frac{s^2}{2}\right)^{k-1} \sum_{\substack{\alpha_i \in \{0,1\} \\ \alpha_1 + \alpha_2 + \dots = 1}} \prod_{m=3}^{\infty} \binom{k}{\alpha_1, \alpha_2, \dots} \left(\frac{s^m}{m!}\right)^{\alpha_m} + \dots \right\} \\ &= \frac{1}{k!} \frac{s^{2k}}{2^k} [(\log |Z|)^2]_{U(N)}^k + s^{2k} O([c_2(N)]^{k-1}). \end{aligned}$$

Here we have used the fact that the second cumulant equals the second moment and that the complicated expression in the second term can be bounded by a constant (this follows from the series expansion of the exponential, but the proof is tedious).

If $n = 2k$ is even, then we have an easy identification with the first expansion, which gives

$$[(\log |Z|)^n]_{U(N)} = \frac{n!}{2^k k!} [(\log |Z|)^2]_{U(N)}^k = (2k-1)!! [(\log |Z|)^2]_{U(N)}^k,$$

as required.

We would expect the error term to be of order $(k-1)$, but in this case we would not get the same power of s - we would have to add at least 3 from the other factors, since we cannot find other cumulants of order 2. So we have to take the third term into

account, from which we can construct the n^{th} power by adding the fourth cumulant. Since $c_2(N) \sim \frac{1}{2} \log N$, we get

$$M_N(s) = \frac{1}{k!} \frac{s^{2k}}{2^k} [(\log |Z|)^2]_{U(N)}^k + s^{2k} O\left(\left(\frac{1}{2} \log N\right)^{k-2}\right)$$

and the first part of the conclusion follows upon differentiating n times.

The other case follows in exactly the same way, but in this case we only take into account the error term of order $k - 1$ from above. $\blacksquare$

3.2. Value distribution of $\text{Im } \log \mathbf{Z}$. This subsection is very similar to the previous one. Most results have a direct correspondent, so will not prove them in full detail this time, but we will write down the key steps where the calculations differ. The central limit theorem for $\text{Im } \log Z$ was first proved by Costin and Lebowitz [18] for matrices in Gaussian ensembles and by Wieand for the CUE case, using a different method [19]. In [18], the authors proved that the eigenvalue counting function is Gaussian in the limit, and the connection comes from ([1])

$$n(U, \theta, \theta') = \frac{(\theta' - \theta)N}{2\pi} + \frac{1}{\pi} \text{Im} \log \frac{Z(U, \theta')}{Z(U, \theta)},$$

where $n(U, \theta, \theta')$ counts the number of eigenangles θ_n with $\theta < \theta_n < \theta'$.

Again, we look at the generating function of $\text{Im } \log Z$, but we scale the argument by $-i$ to get the right averages:

$$\begin{aligned} L_N(-it) &= [e^{-it(\text{Im } \log Z)}]_{U(N)} = [e^{t \text{Im } \log Z}]_{U(N)} \\ &= \sum_{k=0}^{\infty} \frac{t^k [(\text{Im } \log Z)^k]_{U(N)}}{k!}. \end{aligned}$$

The expansion of the logarithm gives the cumulants $d_k(N)$:

$$\log L_N(-it) = \sum_{k=1}^{\infty} \frac{d_k(N) t^k}{k!}$$

and the probability density is given by

$$\sigma_N(x) = [\delta(\text{Im } \log Z - x)]_{U(N)} = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{iyx} e^{-iy \text{Im } \log Z} dy = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-iyx} L_N(y) dy.$$

(the argument of $L_N(y)$ is scaled by $-i$).

As before, we derive an expression for the cumulants:

Lemma 3.8.

$$d_k(N) = \frac{(-i)^n}{2^n} \sum_{k=1}^N [-\psi^{(n-1)}(k) + (-1)^{n-1} \psi^{(n-1)}(k)], \quad \forall k \geq 1.$$

Proof. From Theorem 1.17 we deduce that

$$\log L_N(-it) = \sum_{k=1}^N 2 \log \Gamma(k) - \log \Gamma(k - it/2) - \log \Gamma(k + it/2).$$

Moreover, $d_n(N) = \frac{d^n}{dt^n} \log L_N(-it) \Big|_{t=0}$ and the result in the proof of Lemma 3.1 can be extended to $\alpha \in \mathbb{C}^*$. Setting $\alpha = 1/2$ and $\alpha = -1/2$ this time yields

$$\begin{aligned} d_n(N) &= (-i)^n \frac{d^n}{d(-it)^n} \log L_N(-it) \\ &= (-i)^n \sum_{k=1}^N \left[-\left(\frac{1}{2}\right)^n \psi^{(n-1)}(k) - \left(-\frac{1}{2}\right)^n \psi^{(n-1)}(k) \right] \\ &= \left(\frac{1}{2i}\right)^n \sum_{k=1}^N [-\psi^{(n-1)}(k) + (-1)^{n+1} \psi^{(n-1)}(k)], \end{aligned}$$

as required. ■

Remark 3.9. $d_n(N) = 0$ for n odd and $d_n(N) = \frac{(-1)^{n/2+1}}{2^{n-1}} \sum_{k=1}^N \psi^{(n-1)}(k)$, for n even, which is a significant difference from the real case. However, it is easily seen that

$$d_{2n}(N) = \frac{(-1)^{n+1}}{2^{2n-1} - 1} c_{2n}(N), \quad \forall n \geq 1,$$

so we obtain similar asymptotics ([1]):

$$\begin{aligned} d_2(N) &= c_2(N) = \frac{1}{2} \log N + \frac{1}{2}(\gamma + 1) + \frac{1}{24N^2} - \frac{1}{80N^4} + O\left(\frac{1}{N^6}\right), \\ d_{2n}(N) &= \frac{(-1)^{n+1}}{2^{2n-1} - 1} \left(\Gamma(2n) \zeta(2n-1) - \frac{(2n-3)!}{N^{2n-2}} \right) + O(N^{1-2n}), \quad \forall n > 1. \end{aligned}$$

As before, consider the scaled probability density, which is obtained by a change of variable $y = d_2(N)x$:

$$\tilde{\sigma}_N(x) = \sqrt{d_2(N)} \sigma_N\left(\sqrt{d_2(N)}x\right).$$

We can apply the exact same steps as in the proof of 3.5 to get

$$\begin{aligned} \tilde{\sigma}_N(x) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \exp\{-iyx + \log L_N(-i(iy))\} dy \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \exp\left(-iyx + \sum_{k=1}^{\infty} \frac{d_k(N)(iy)^k}{k!}\right) dy \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \exp\left(-iyx - \frac{y^2}{2} + \sum_{k=2}^{\infty} (-1)^k \frac{d_{2k}(N)y^{2k}}{d_2(N)^k k!}\right) dy \\ &= \frac{1}{\sqrt{2\pi}} e^{-x^2/2} + \frac{1}{2\pi} \int_{-\infty}^{\infty} \exp\left(-iyx - \frac{y^2}{2}\right) \left(\sum_{k=2}^{\infty} \frac{D_{2k}y^{2k}}{d_2(N)^k}\right) dy, \end{aligned}$$

where the coefficients $D_{2n}(N)$ are defined in the same way as $C_n(N)$, for $n \geq 1$. Also

$$\tilde{\sigma}_N(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} + \frac{1}{\sqrt{2\pi}} \sum_{n=2}^{\infty} \frac{D_{2n}}{d_2(N)^n} e^{-x^2/2} (-ix)^p \sum_{p=0}^{2n} \binom{2n}{p} (2n-p-1)!!,$$

for $2n-p$ even and 0 for $2n-p$ odd, where we have used the corresponding result for $\text{Re} \log Z$ and $i^{2n-p} = (-i)^p (-1)^n$. Since all the odd cumulants are 0, $\tilde{\sigma}_N(x)$ is

an even function of order $(\log N)^{-2}$, which is asymptotically smaller than in the real setting.

The last result of the subsection is concerned with finding estimates for the moments of $\text{Im } \log Z$. Since the odd cumulants vanish, so do the odd moments (which can be expressed in terms of the cumulants - see [this section](#)). In this case we have

Proposition 3.10.

$$\left[(\text{Im } \log Z)^{2k} \right]_{U(N)} = (2k-1)!! \left[(\text{Im } \log Z)^2 \right]_{U(N)}^k + O((\log N)^{k-2}),$$

for all $k \geq 1$.

3.3. Independence. The last step of the proof of [1.11](#) is to show that the two densities shown to converge to $N(0, 1)$ also converge independently to the joint normal distribution. We start by computing the generating functions, then we inspect the joint value distribution and rewrite it in a way that highlights the limiting Gaussian distribution.

As in the second section, we use the Weyl Integration Formula to compute the joint probability density, which is just the product of the previous two.

Theorem 3.11.

$$Z(t, s) := \left[e^{t \text{Re } \log Z + i s \text{Im } \log Z} \right]_{U(N)} = \prod_{k=1}^N \frac{\Gamma(k) \Gamma(t+k)}{\Gamma(k+t/2+s/2) \Gamma(k+t/2-s/2)},$$

for every $s, t \in \mathbb{R}$ for which the average is defined.

Proof.

$$\begin{aligned} Z(t, s) &= \\ &= \frac{1}{(2\pi)^N N!} \int_{[0, 2\pi]^N} \prod_{1 \leq j < k \leq N} |e^{i\theta_j} - e^{i\theta_k}|^2 \prod_{n=1}^N |1 - e^{i(\theta_n - \theta)}|^t \\ &\quad \prod_{l=1}^N \exp \left(-i s \sum_{m=1}^{\infty} \frac{\sin[(\theta_l - \theta)m]}{m} \right) d\theta_1 \dots d\theta_N = \\ &= \frac{1}{(2\pi)^N N!} \int_{[0, 2\pi]^N} \prod_{1 \leq j < k \leq N} |e^{i\theta_j} - e^{i\theta_k}|^2 \prod_{n=1}^N |1 - e^{i\theta_n}|^t \prod_{l=1}^N \exp \left(-i s \sum_{m=1}^{\infty} \frac{\sin(\theta_l m)}{m} \right) \\ &\quad d\theta_1 \dots d\theta_N = \\ &= \frac{2^{N(N-1)} 2^{tN}}{(2\pi)^N N!} \int_{[0, 2\pi]^N} \prod_{1 \leq j < k \leq N} \left| \sin \frac{\theta_j - \theta_k}{2} \right|^2 \prod_{n=1}^N \left| \sin \frac{\theta_n}{2} \right|^t \prod_{n=1}^N \exp \left(-\frac{i s (\pi - \theta_n)}{2} \right) \\ &\quad d\theta_1 \dots d\theta_N = \\ &= \frac{2^{N^2} 2^{tN}}{N! (2\pi)^N} \int_{[-\pi/2, \pi/2]^N} \prod_{1 \leq j < k \leq N} |\tan \psi_j - \tan \psi_k|^2 \prod_{n=1}^N (\cos^2 \psi_n)^{N-1} \prod_{n=1}^N |\cos \psi_n|^t \times \end{aligned}$$

$$\begin{aligned}
& \times \prod_{k=1}^N (\cos \psi_k + i \sin \psi_k)^s d\psi_1 \dots d\psi_N = \\
& = \frac{2^{N^2} 2^{tN}}{N! (2\pi)^N} \int_{\mathbb{R}^N} \prod_{1 \leq j < k \leq N} |x_j - x_k|^2 \prod_{n=1}^N \left(\frac{1}{1 + x_n^2} \right)^{N+t/2} \prod_{l=1}^N \left(\frac{1 + ix_l}{\sqrt{1 + x_l^2}} \right)^s dx_1 \dots dx_N \\
& = \frac{2^{N^2} 2^{tN}}{N! (2\pi)^N} \int_{\mathbb{R}^N} \prod_{1 \leq j < k \leq N} |x_j - x_k|^2 \prod_{n=1}^N (1 + ix_n)^{-N-t/2+s/2} (1 - ix_n)^{-N-t/2-s/2} \\
& \quad dx_1 \dots dx_N = \\
& = \frac{2^{N^2} 2^{tN}}{N! (2\pi)^N} J(1, 1, N + t/2 - s/2, N + t/2 + s/2, 1, N),
\end{aligned}$$

i.e. $Z(t, s)$ is a Selberg integral with $a = b = 1$, $\alpha = N + t/2 - s/2$, $\beta = N + t/2 + s/2$ and $\gamma = 1$. Here we have used [A.1](#) again and the same substitutions as in Section 2. It is easy to check that the conditions of Theorem [2.1](#) are all fulfilled if $t/2 \pm s/2 > -1$ and $t > -1$. In this case the explicit formula for $Z(t, s)$ is

$$\begin{aligned}
Z(t, s) &= \frac{2^{N^2} 2^{tN}}{N! (2\pi)^N} \frac{(2\pi)^N}{2^{(2N+t)N - N(N-1) - N}} \\
& \quad \prod_{k=0}^{N-1} \frac{\Gamma(2+k) \Gamma(2N+t - (N+k-1) - 1)}{\Gamma(2) \Gamma(N+t/2 - s/2 - k) \Gamma(N+t/2 + s/2 + k)} \\
&= \frac{1}{N!} \prod_{\substack{j=N-k \\ 1 \leq j \leq N}} \frac{\Gamma(2+N-j) \Gamma(t+j)}{\Gamma(j+t/2 - s/2) \Gamma(j+t/2 + s/2)} \\
&= \prod_{j=1}^N \frac{\Gamma(j) \Gamma(t+j)}{\Gamma(j+t/2 - s/2) \Gamma(j+t/2 + s/2)},
\end{aligned}$$

since $\frac{1}{N!} \prod_{j=1}^N \Gamma(2+N-j) = \prod_{j=1}^N \Gamma(j)$ by basic properties of the gamma function. $\blacksquare$

The joint probability density that we need is

$$\tau_N(x, y) = [\delta(\log |Z| - x) \delta(\operatorname{Im} \log Z - y)]_{U(N)},$$

and our next theorem will provide the last part of the proof of Theorem [1.11](#).

Theorem 3.12.

$$\lim_{N \rightarrow \infty} \tilde{\tau}_N(x, y) = \frac{1}{2\pi} \exp\left(-\frac{x^2}{2}\right) \exp\left(-\frac{y^2}{2}\right),$$

where $\tilde{\tau}_N(x, y) := \sqrt{c_2(N) d_2(N)} \tau_N\left(\sqrt{c_2(N)} x, \sqrt{d_2(N)} y\right)$ is the scaled joint distribution.

Before proceeding to the proof, we would like to write $Z(t, s)$ in a different way. Specifically, we can use 3.11 and the Taylor expansion for two variables to get

$$\begin{aligned} \log Z(t, s) &= \sum_{j=1}^N \log \Gamma(j) + \log \Gamma(t + j) - \log \Gamma(j + t/2 + s/2) - \log \Gamma(j + t/2 - s/2) \\ &= \sum_{n=0}^{\infty} \sum_{\substack{i+j=n \\ i \geq 0, j \geq 0}} \frac{\partial^{i+j}}{\partial t^i \partial s^j} \log Z(t, s) \Big|_{(0,0)} t^i s^j \stackrel{\text{not}}{=} \sum_{n=0}^{\infty} \sum_{\substack{i+j=n \\ i \geq 0, j \geq 0}} \alpha_{ij} t^i s^j. \end{aligned}$$

The expansion is valid, since the gamma function is infinitely differentiable where it is defined. Clearly $\alpha_{00} = 0$, since $Z(0, 0) = 1$. By letting $t = 0$ or $s = 0$, we recover the initial expansion of the logarithm of the corresponding generating function, which has cumulant coefficients that can be derived as in 1.7. For instance,

$$Z(0, t) = [e^{t \operatorname{Re} \log Z}]_{U(N)} = \alpha_{10} t + \alpha_{20} \frac{t^2}{2!} + \alpha_{30} \frac{t^3}{3!} + \dots,$$

so clearly $\alpha_{n0} = c_n(N)$, for $n \neq 0$. In the same way we get $\alpha_{0n} = i^n d_n(N)$, where i^n comes from the expansion of $[e^{is \operatorname{Im} \log Z}]_{U(N)}$.

We also have

$$\begin{aligned} \alpha_{mn} &= \frac{\partial^{m+n}}{\partial t^m \partial s^n} \log Z(t, s) \Big|_{(0,0)} \\ &= \frac{\partial^m}{\partial t^m} \left[\sum_{j=1}^N \frac{1}{2^n} \{ -\psi^{(n-1)}(j + t/2 + s/2) + (-1)^{n-1} \psi^{(n-1)}(j + t/2 - s/2) \} \right]_{(0,0)} \\ &= \sum_{j=1}^N \frac{1}{2^{m+n}} [-\psi^{(n+m-1)}(j + t/2 + s/2) + (-1)^{n-1} \psi^{(n+m-1)}(j + t/2 - s/2)]_{(0,0)} \\ &= \begin{cases} -\frac{1}{2^{n+m-1}} \sum_{j=1}^N \psi^{(n+m-1)}(j) & \text{if } n \text{ odd} \\ 0 & \text{if } n \text{ even.} \end{cases} \end{aligned}$$

Proof of Theorem 3.12:

From the above, the joint value distribution can be written as

$$\begin{aligned} \tau_N(x, y) &= \frac{1}{4\pi^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-itx - isy} [e^{it \log |Z|} e^{is \operatorname{Im} \log Z}]_{U(N)} dt ds \\ &= \frac{1}{4\pi^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-itx - isy} \prod_{j=1}^N \frac{\Gamma(j) \Gamma(it + j)}{\Gamma(j + it/2 + s/2) \Gamma(j + it/2 - s/2)} dt ds \\ &= \frac{1}{4\pi^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-itx - isy} \exp \left(\sum_{n=1}^{\infty} \sum_{\substack{i+j=n \\ i \geq 0, j \geq 0}} \alpha_{ij} t^i s^j \right) dt ds. \end{aligned}$$

The generating function is again computed with the Selberg Formula (the conditions are still satisfied for $s \in (-2, 2)$ and $N \geq 2$). Note that $\alpha_{10} = \alpha_{01} = \alpha_{11} = 0$ and also $\alpha_{20} = -\alpha_{02} = \frac{1}{2} \log N + \frac{1}{2}(\gamma + 1) + \frac{1}{24N^2} - \frac{1}{80N^4} + O\left(\frac{1}{N^6}\right)$.

Lemma 3.13. $\alpha_{mn} = O(1)$, for $m + n \geq 3$.

Proof. Use Proposition 3.3 (and the previous results) to get

$$\begin{aligned}\alpha_{mn} &= -\frac{1}{2^{m+n-1}} \left[\Gamma(m+n) \zeta(m+n-1) - \frac{(m+n-3)!}{N^{m+n-2}} \right] + O\left(\frac{1}{N^{m+n-1}}\right) \\ &= -\frac{(m+n-3)!}{2^{m+n-1}} \left[(m+n-1)(m+n-2) \zeta(m+n-1) - \frac{1}{N^{m+n-2}} \right] \\ &\quad + O\left(\frac{1}{N^{m+n-1}}\right) = O(1),\end{aligned}$$

since $\zeta(m+n-1) \rightarrow 1$ and $\frac{1}{N^{m+n-2}} \rightarrow 0$ if one of the m, n goes to ∞ . ■

Consider now the scaled joint distribution

$$\tilde{\tau}_N(x, y) = \sqrt{c_2(n)d_2(N)} \tau_N\left(\sqrt{c_2(N)}x, \sqrt{d_2(N)}y\right).$$

This satisfies

$$\begin{aligned}\tilde{\tau}_N(x, y) &= \frac{1}{4\pi^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp\left(-itx - isy - \frac{t^2}{2} - \frac{s^2}{2}\right. \\ &\quad \left.+ \frac{\alpha_{30}}{\alpha_{20}^{3/2} 3!} (it)^3 + \frac{\alpha_{21}}{\alpha_{20}^{3/2} 2!} (it)^2 s + \frac{\alpha_{12}}{\alpha_{20}^{3/2} 2!} its^2 + \frac{\alpha_{03}}{\alpha_{20}^{3/2} 3!} s^3 + \dots\right) dt du \\ &= \frac{1}{4\pi^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp\left(-itx - isy - \frac{t^2}{2} - \frac{s^2}{2}\right) \left(1 + O\left((\log N)^{-3/2}\right)\right) dt du,\end{aligned}$$

because, by 3.13, every cumulant in the denominator is $O(1)$ and the others are asymptotic to $\frac{1}{2} \log N$ (the constant gets absorbed in $O(1)$). This can be split into two terms, and the first is the product of two Gaussian integrals, computed before:

$$\begin{aligned}\frac{1}{4\pi^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp\left(-itx - isy - \frac{t^2}{2} - \frac{s^2}{2}\right) &= \frac{1}{4\pi^2} \int_{-\infty}^{\infty} e^{-itx - t^2/2} dt \int_{-\infty}^{\infty} e^{-isy - s^2/2} ds \\ &= \frac{1}{2\pi} \exp\left(-\frac{x^2}{2}\right) \exp\left(-\frac{y^2}{2}\right),\end{aligned}$$

which follows immediately from the proof of Theorem 3.6. The other integral is simply $\frac{1}{2\pi} \exp\left(-\frac{x^2}{2}\right) \exp\left(-\frac{y^2}{2}\right) \times O\left((\log N)^{-3/2}\right)$, which clearly vanishes as $N \rightarrow \infty$. So

$$\lim_{N \rightarrow \infty} \tilde{\tau}_N(x, y) = \frac{1}{2\pi} \exp\left(-\frac{x^2}{2}\right) \exp\left(-\frac{y^2}{2}\right),$$

which is separable, so indeed the real and imaginary parts of $\log Z$ converge independently to the probability density of $N(0, 1)$ and Theorem 1.11 is proved. ■

4. ASYMPTOTIC BEHAVIOUR OF THE GENERATING FUNCTIONS

In this section, we will try to deepen our understanding about the generating functions $M_N(s)$ and $L_N(s)$ and the value distributions of $\log Z$ for various ensembles by looking at the leading order asymptotics. The previous results were mostly derived for the CUE case, but the techniques can be adapted to the COE and CSE case and also to the compact matrix groups $SO(2n)$ and $USp(2n)$ (see [15]).

4.1. Other cumulants. We start off by computing the cumulants for the COE and CSE. We can compute the generating functions as in the second section, but with adjusted coefficients ($\alpha = 1$ for the COE, $\alpha = 2$ for the CUE and $\alpha = 4$ for the CSE):

Theorem 4.1. *We have ([1])*

$$\begin{aligned} M_N(s) &= [e^{s \operatorname{Re} \log Z}]_{C^*E} = \\ &= \frac{[(\alpha/2)!]^N}{(N\alpha/2)!(2\pi)^N} \int_{[0,2\pi]^N} \prod_{1 \leq j < k \leq N} |e^{i\theta_j} - e^{i\theta_k}|^\alpha \left| \prod_{n=1}^N (1 - e^{i(\theta_n - \theta)}) \right|^s d\theta_1 \dots d\theta_N \\ &= \prod_{k=0}^N \frac{\Gamma(1 + k\alpha/2) \Gamma(1 + s + k\alpha/2)}{(\Gamma(1 + s/2 + k\alpha/2))^2}. \end{aligned}$$

$$\begin{aligned} L_N(s) &= [e^{is \operatorname{Im} \log Z}]_{C^*E} = \\ &= \frac{[(\alpha/2)!]^N}{(N\alpha/2)!(2\pi)^N} \int_{[0,2\pi]^N} \prod_{1 \leq j < k \leq N} |e^{i\theta_j} - e^{i\theta_k}|^\alpha \prod_{n=1}^N \exp \left(-is \frac{\alpha}{2} \sum_{k=1}^{\infty} \frac{\sin [(\theta_n - \theta)k]}{k} \right) \\ &\quad d\theta_1 \dots d\theta_N \\ &= \prod_{k=0}^{N-1} \frac{(\Gamma(1 + k\alpha/2))^2}{\Gamma(k\alpha/2 + 1 + s/2) \Gamma(k\alpha/2 + 1 - s/2)}, \end{aligned}$$

where the expressions are obtained from Weyl's Integration Formula.

Lemma 3.1 can be generalised:

Lemma 4.2. *The n^{th} cumulant satisfies*

$$c_n^\alpha(N) = \frac{2^{n-1} - 1}{2^{n-1}} \sum_{k=0}^{N-1} \psi^{(n-1)}(1 + k\alpha/2).$$

Similarly, from 4.1:

Lemma 4.3.

$$d_n^\alpha(N) = \begin{cases} -\frac{(-1)^{n/2+1}}{2^{n-1}} \sum_{k=0}^{N-1} \psi^{(n-1)}(1 + j\alpha/2) & \text{if } n \text{ even} \\ 0 & \text{if } n \text{ odd.} \end{cases}$$

Remark 4.4. For $n = 2$, it is easily seen that $c_2^\alpha(N) = d_2^\alpha(N)$.

We use the same ideas as in Section 3 to compute the cumulants. First, we have:

Proposition 4.5.

$$\lim_{N \rightarrow \infty} c_n^1(N) = (2^{n-1} - 1)(-1)^n \Gamma(n) \left(\zeta(n-1) - \frac{2^n - 1}{2^n} \zeta(n) \right) \text{ and}$$

$$\lim_{N \rightarrow \infty} c_n^4(N) = \frac{2^{n-1} - 1}{2^n} (-1)^n \Gamma(n) \left(\zeta(n-1) + \frac{2^n - 1}{2^n} \zeta(n) \right).$$

Proof. Let us look at the limiting value of the general cumulant $c_n^\alpha(\infty) = \lim_{N \rightarrow \infty} c_n^\alpha(N)$. By Lemma 4.2 and the integral representation of polygamma functions,

$$\begin{aligned} c_n^\alpha(\infty) &= \frac{2^{n-1} - 1}{2^{n-1}} \lim_{N \rightarrow \infty} \sum_{k=0}^N (-1)^n \int_0^\infty \frac{t^{n-1} e^{-(1+k\alpha/2)t}}{1 - e^{-t}} dt \\ &= \frac{2^{n-1} - 1}{2^{n-1}} (-1)^n \int_0^\infty \left(e^{-t} t^{n-1} \sum_{m=0}^\infty e^{-mt} \sum_{k=0}^\infty e^{-\alpha k t/2} \right) dt \\ &= \frac{2^{n-1} - 1}{2^{n-1}} (-1)^n \sum_{k=0}^\infty \sum_{m=0}^\infty \int_0^\infty e^{-(1+m+\alpha k/2)t} t^{n-1} dt \\ &= \frac{2^{n-1} - 1}{2^{n-1}} (-1)^n \sum_{k=0}^\infty \sum_{m=1}^\infty \Gamma(n) (m + \alpha k/2)^{-n}. \end{aligned}$$

To prove the last equality, let $I(k, m, n) = \int_0^\infty e^{-(1+m+\alpha k/2)t} t^{n-1} dt$, for $k, m \geq 0$. Integrating by parts,

$$\begin{aligned} I(k, m, n) &= -\frac{1}{1+m+\alpha k/2} \int_0^\infty (e^{-(1+m+\alpha k/2)t})' t^{n-1} dt \\ &= -\frac{1}{1+m+\alpha k/2} [e^{-(1+m+\alpha k/2)t} t^{n-1}]_0^\infty \\ &\quad + \frac{1}{1+m+\alpha k/2} (n-1) \int_0^\infty e^{-(1+m+\alpha k/2)t} t^{n-2} dt \\ &= \frac{n-1}{1+m+\alpha k/2} I(k, m, n-1). \end{aligned}$$

By induction we get

$$I(k, m, n) = \frac{(n-1)!}{(1+m+\alpha k/2)^{n-1}} I(k, m, 1) = \frac{(n-1)!}{(1+m+\alpha k/2)^n},$$

so we have derived the general expression.

Observe that we can reduce the double sum to a single variable p by computing the number of ways in which $m + \alpha k/2 = p$, for $p \geq 1$, without omitting any pair (m, k) . As p spans $\mathbb{N}$, so do m, k and we recover each pair (but they will be scattered).

If $\alpha = 1$, there are p of them for k even. If k is odd, we have to take p a half-integer, say $p = q/2$, for $q \geq 3$ odd. In this case we get $(q-1)/2$ possibilities.

If $\alpha = 4$, $m + 2k = p$ and we have $p/2$ ways if p even, $(p+1)/2$ ways if p odd. So

$$c_n^1(\infty) = \frac{2^{n-1} - 1}{2^{n-1}} (-1)^n \Gamma(n) \left(\sum_{p=1}^\infty \frac{p}{p^n} + \sum_{\substack{q=3 \\ q \text{ odd}}}^\infty \frac{q-1}{2(q/2)^n} \right)$$

$$\begin{aligned}
&= (2^{n-1} - 1)(-1)^n \Gamma(n) \left(\frac{1}{2^{n-1}} \zeta(n-1) + \sum_{p=1}^{\infty} \frac{2p+1}{(2p+1)^n} - \sum_{p=1}^{\infty} \frac{1}{(2p+1)^n} \right) \\
&= (2^{n-1} - 1)(-1)^n \Gamma(n) \left(\frac{1}{2^{n-1}} \zeta(n-1) + \sum_{p=1}^{\infty} \frac{1}{p^{n-1}} - \frac{1}{2^{n-1}} \sum_{p=1}^{\infty} \frac{1}{p^{n-1}} - \right. \\
&\quad \left. - \sum_{p=1}^{\infty} \frac{1}{p^n} + \frac{1}{2^n} \sum_{p=1}^{\infty} \frac{1}{p^n} \right) \\
&= (2^{n-1} - 1)(-1)^n \Gamma(n) \left(\zeta(n-1) - \left(1 - \frac{1}{2^n}\right) \zeta(n) \right)
\end{aligned}$$

and

$$\begin{aligned}
c_n^4(\infty) &= \frac{2^{n-1} - 1}{2^{n-1}} (-1)^n \Gamma(n) \left(\sum_{k=1}^{\infty} \frac{k}{(2k-1)^n} + \sum_{k=1}^{\infty} \frac{k}{(2k)^n} \right) \\
&= \frac{2^{n-1} - 1}{2^{n-1}} (-1)^n \Gamma(n) \frac{1}{2} \left(\sum_{k=1}^{\infty} \frac{2k-1}{(2k-1)^n} + \sum_{k=1}^{\infty} \frac{1}{(2k-1)^n} + \sum_{k=1}^{\infty} \frac{2k}{(2k)^n} \right) \\
&= \frac{2^{n-1} - 1}{2^n} (-1)^n \Gamma(n) \left(\zeta(n-1) + \left(1 - \frac{1}{2^n}\right) \zeta(n) \right),
\end{aligned}$$

by joining the first and third sum and completing the second one. ■

Remark 4.6. The limit $c_n^\alpha(\infty)$ provides another way to compute the second CUE cumulant. In this case we have to compute the number of ways in which $m+k=p$, for $m, p \geq 1, k \geq 0$. There are p possibilities, one for each $m \in \{1, 2, \dots, p\}$. So

$$c_n^2(\infty) = \frac{2^{n-1} - 1}{2^{n-1}} (-1)^n \Gamma(n) \left(\sum_{p=1}^{\infty} \frac{p}{p^n} \right) = \frac{2^{n-1} - 1}{2^{n-1}} (-1)^n \Gamma(n) \zeta(n-1),$$

which is in accordance with the stronger result 3.3.

Remark 4.7. We can compute the corresponding limiting values for $\text{Im} \log Z$, $d_{2n}^\alpha(\infty) = \lim_{N \rightarrow \infty} d_{2n}^\alpha(N)$, in exactly the same way, since the expressions for the n^{th} cumulant differ only by a constant. From 4.2 and 4.3,

$$\begin{aligned}
d_{2n}^1(\infty) &= (-1)^{3n/2+1} \Gamma(n) \left(\zeta(n-1) - \left(1 - \frac{1}{2^n}\right) \zeta(n) \right) \\
d_{2n}^4(\infty) &= \frac{(-1)^{3n/2+1}}{2^n} \Gamma(n) \left(\zeta(n-1) + \left(1 - \frac{1}{2^n}\right) \zeta(n) \right),
\end{aligned}$$

for every $n \geq 1$.

Before we proceed to compute the leading-order asymptotics, we analyse the second cumulants for each ensemble. First, recall that

$$c_n^\alpha(N) = \frac{2^{n-1} - 1}{2^{n-1}} \sum_{k=0}^{N-1} \psi^{(n-1)}(1 + k\alpha/2).$$

We will also need the recurrence formula for the polygamma function, for $\text{Re}(z) > 0$:

$$\psi^{(1)}(z+1) = \psi^{(1)}(z) - \frac{1}{z^2},$$

which is proved in [A.7](#).

COE second cumulant. Since $\alpha = 1$, we will work with half-integer-valued polygamma functions and we have to check the parity of N separately. For N even,

$$\begin{aligned}
 c_2^1(N) &= \frac{1}{2} \sum_{j=0}^{N-1} \psi^{(1)}(1 + j/2) \\
 &= \frac{1}{2} \left\{ \left(\frac{N}{2} \psi^{(1)}(1) - \sum_{k=1}^{(N-2)/2} \frac{N/2 - k}{k^2} \right) + \left(\frac{N}{2} \psi^{(1)}(1/2) - \sum_{k=1}^{N/2} \frac{4(N/2 - k + 1)}{(2k - 1)^2} \right) \right\} \\
 &= \frac{1}{2} \left\{ 2N\zeta(2) - \left(\frac{N}{2} \sum_{k=1}^{(N-2)/2} \frac{1}{k^2} + 2N \sum_{k=1}^{N/2} \frac{1}{(2k - 1)^2} \right) + \sum_{k=1}^{(N-2)/2} \frac{1}{k} + \sum_{k=1}^{N/2} \frac{2}{2k - 1} \right. \\
 &\quad \left. - 2 \sum_{k=1}^{N/2} \frac{1}{(2k - 1)^2} \right\} = \\
 &= \frac{1}{2} \left\{ 2N\zeta(2) - 2N \sum_{k=1}^{N-1} \frac{1}{k^2} - 2 \sum_{k=1}^{N/2} \frac{1}{(2k - 1)^2} + 2 \sum_{k=1}^{N-1} \frac{1}{k} \right\} \\
 &= N \left(\zeta(2) - \sum_{k=1}^{N-1} \frac{1}{k^2} \right) + \sum_{k=1}^{N-1} \frac{1}{k} - \sum_{k=1}^{N-1} \frac{1}{k^2} + \frac{1}{4} \sum_{k=1}^{(N-2)/2} \frac{1}{k^2},
 \end{aligned}$$

where we have used $\psi^{(1)}(1) = \zeta(2)$, $\psi^{(1)}(1/2) = 3\zeta(2)$ (results that follow from [A.9](#))

and the identities $2 \sum_{k=1}^{N/2} \frac{1}{2k - 1} = 2 \sum_{k=1}^{N-1} \frac{1}{k} - \sum_{k=1}^{(N-2)/2} \frac{1}{k}$ and $\sum_{k=1}^{N/2} \frac{1}{(2k - 1)^2} = \sum_{k=1}^{N-1} \frac{1}{k^2} - \frac{1}{4} \sum_{k=1}^{(N-2)/2} \frac{1}{k^2}$. From the Euler-Maclaurin formula, $\sum_{k=1}^n \frac{1}{k^2} \sim \zeta(2) - \frac{1}{n} + O(n^{-2})$ and $\sum_{k=1}^n \frac{1}{k} = \log n + \gamma + O(n^{-1})$, for $n \geq 1$, so we end up with

$$\begin{aligned}
 c_2^1(N) &= N \left[\frac{1}{N-1} + O(N^{-2}) \right] + \log N + \gamma - \frac{3}{4}\zeta(2) + \frac{1}{N-1} - \frac{1}{2(N-2)} + O(N^{-1}) \\
 &= \log N + 1 + \gamma - \frac{3}{4}\zeta(2) + O(N^{-1}).
 \end{aligned}$$

For N odd, the calculations are very similar and we get the same result by adjusting a few terms:

$$\begin{aligned}
 c_2^1(N) &= \frac{1}{2} \sum_{j=0}^{N-1} \psi^{(1)}(1 + j/2) \\
 &= \frac{1}{2} \left\{ \frac{N+1}{2} (\psi^{(1)}(1) + \psi^{(1)}(1/2)) - \sum_{k=1}^{(N-1)/2} \frac{(N+1)/2 - k}{k^2} - \right.
 \end{aligned}$$

$$\begin{aligned}
& - \sum_{k=1}^{(N-1)/2} \frac{(N+3)/2 - k}{(k - 1/2)^2} \Bigg\} = \\
& = (N+1) \left(\zeta(2) - \sum_{k=1}^{\infty} \frac{1}{k^2} \right) + \sum_{k=1}^{N-1} \frac{1}{k} + \sum_{k=1}^{N-1} \frac{1}{k^2} - \frac{1}{4} \sum_{k=1}^{(N-1)/2} \frac{1}{k^2} \\
& = \log N + 1 + \gamma - \frac{3}{4} \zeta(2) + O(N^{-1}).
\end{aligned}$$

CSE second cumulant. This is easier to derive than in the COE case, since we are only dealing with integer-valued polygamma functions. So

$$\begin{aligned}
c_2^4(N) &= \frac{1}{2} \sum_{j=0}^{N-1} \psi^{(1)}(1+2j) = \frac{1}{2} \left[\psi^{(1)}(1) + \sum_{j=1}^{N-1} \left(\psi^{(1)}(1) - \sum_{m=1}^{2j} \frac{1}{m^2} \right) \right] = \\
&= \frac{N}{2} \psi^{(1)}(1) - \frac{1}{2} \sum_{k=1}^{N-1} \frac{N-k}{(2k-1)^2} - \frac{1}{2} \sum_{k=1}^{N-1} \frac{N-k}{(2k)^2} \\
&= \frac{N}{2} \zeta(2) - \frac{N}{2} \sum_{k=1}^{N-1} \frac{1}{(2k-1)^2} + \frac{1}{4} \sum_{k=1}^{N-1} \frac{2k-1}{(2k-1)^2} + \frac{1}{4} \sum_{k=1}^{N-1} \frac{1}{(2k-1)^2} \\
&\quad - \frac{N}{8} \sum_{k=1}^{N-1} \frac{1}{k^2} + \frac{1}{8} \sum_{k=1}^{N-1} \frac{1}{k} = \\
&= \frac{N}{2} \zeta(2) - \frac{N}{2} \sum_{k=1}^{2N-2} \frac{1}{k^2} + \frac{N}{8} \sum_{k=1}^{N-1} \frac{1}{k^2} + \frac{1}{4} \sum_{k=1}^{2N-2} \frac{1}{k^2} - \frac{1}{16} \sum_{k=1}^{N-1} \frac{1}{k^2} \\
&\quad - \frac{N}{8} \sum_{k=1}^{N-1} \frac{1}{k^2} + \frac{1}{8} \sum_{k=1}^{2N-2} \frac{1}{k} = \\
&= \frac{N}{2} \left(\zeta(2) - \sum_{k=1}^{2N-2} \frac{1}{k^2} \right) + \left(\frac{1}{4} \sum_{k=1}^{2N-2} \frac{1}{k^2} - \frac{1}{16} \sum_{k=1}^{N-1} \frac{1}{k^2} \right) \\
&\quad + \frac{1}{4} (\log 2 + \log N + \gamma) + O(N^{-1}) = \\
&= \frac{N}{2} \left(\zeta(2) - \sum_{k=1}^{2N-2} \frac{1}{k^2} \right) + \frac{3}{16} \zeta(2) - \frac{1}{4} \sum_{k=2N-1}^{\infty} \frac{1}{k^2} + \frac{1}{16} \sum_{k=N}^{\infty} \frac{1}{k^2} \\
&\quad + \frac{1}{4} (\log 2 + \log N + \gamma) + O(N^{-1}).
\end{aligned}$$

From the Euler-Maclaurin formula, $\sum_{k=1}^{2N-2} \frac{1}{k^2} \sim \zeta(2) - \frac{1}{2N-2} + O(N^{-2})$, so the first term is $1/4 + O(N^{-1})$. The two infinite sums are also $O(N^{-1})$, since we can use the inequalities $\frac{1}{k(k+1)} \leq \frac{1}{k^2} \leq \frac{1}{k(k-1)}$, for every $k \geq 2$ and the identity $\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$. We end up with

$$c_2^4(N) = \frac{1}{4} (\log N + \log 2 + \gamma + 1) + \frac{3}{16} \zeta(2) - \frac{1}{4(2N-1)} + \frac{1}{16(N-1)} + O(N^{-1})$$

$$= \frac{1}{4} \log N + \frac{1}{4}(\gamma + 1) + \frac{1}{4} \log 2 + \frac{3}{16} \zeta(2) + O(N^{-1}).$$

Remark 4.8. By 4.4, we now have the second cumulants for both real and imaginary parts of $\log Z(U, \theta)$.

4.2. Leading order asymptotics. We turn now to the proof of Theorem 1.14 and we state and prove similar theorems for the COE and the CSE. But before we proceed, we need the definition of the Barnes G-function (see also [20], [21]):

$$G(1+z) = (2\pi)^{z/2} \exp\left(-\frac{(1+\gamma)z^2+z}{2}\right) \prod_{n=1}^{\infty} \left[\left(1 + \frac{z}{n}\right)^n \exp\left(-z + \frac{z^2}{2n}\right) \right],$$

for every $z \in \mathbb{C}$, which has the nice properties

$$\begin{aligned} G(1) &= 1 \\ G(z+1) &= \Gamma(z)G(z), \text{ for } z \neq 0. \end{aligned}$$

Remark 4.9. The G-function is an entire function of order two, with zeros at the negative integers $n \leq -1$.

Sketch of the proof. The order of an entire function $f(z)$ can be described as the infimum of all n for which $f(z) = O(\exp|z|^n)$ as $z \rightarrow \infty$. The definition of G contains exponentials up to $\exp(z^2)$ and other asymptotically smaller terms: $\left(1 + \frac{z}{n}\right)^n \rightarrow e^z$ (which is cancelled by e^{-z} in the next factor) and $\exp\left(\frac{z^2}{2n}\right) \rightarrow 1$ as $n \rightarrow \infty$. By the triangle inequality and noting that we already have $O(\exp(z^2))$ factors, it follows easily that $G(1+z) = O(\exp|z|^2)$, so it has order 2.

The other property follows by the functional equation for the gamma function, $\Gamma(z+1) = z\Gamma(z)$, for $\text{Re}(z) > 0$. This leads to $\Gamma(s) = \frac{\Gamma(s+n)}{s(s+1)\dots(s+n)}$, which defines a meromorphic continuation of $\Gamma(s)$, for $\text{Re}(s) > -n$, with the only poles at $s \in \{0, -1, \dots, -(n-1)\}$, for any $n \in \mathbb{Z}_{>0}$. Since $G(z) = G(z+1)/\Gamma(z)$, $G(z) = 0$ at the poles, which are all negative integers. However, we have not been able to show that these are indeed all the zeros of the G-function.

We can also prove the following:

Lemma 4.10.

$$\log G(1+z) = (\log(2\pi) - 1) \frac{z}{2} - (1+\gamma) \frac{z^2}{2} + \sum_{n=3}^{\infty} (-1)^{n-1} \zeta(n-1) \frac{z^n}{n}.$$

Proof. Taking logs on both sides and using the expansion $\log(1+z) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{z^n}{n}$, valid for $|z| < 1$ (the series has unit radius of convergence), we get:

$$\begin{aligned} \log G(1+z) &= (\log(2\pi) - 1) \frac{z}{2} - (1+\gamma) \frac{z^2}{2} + \sum_{n=1}^{\infty} n \log\left(1 + \frac{z}{n}\right) - z + \frac{z^2}{2n} \\ &= (\log(2\pi) - 1) \frac{z}{2} - (1+\gamma) \frac{z^2}{2} + n \left(\frac{z}{n} - \frac{z^2}{2n^2} + \frac{z^3}{3n^3} - \dots \right) - z + \frac{z^2}{2n} \end{aligned}$$

$$\begin{aligned}
&= (\log(2\pi) - 1) \frac{z}{2} - (1 + \gamma) \frac{z^2}{2} + \sum_{n=3}^{\infty} (-1)^{n-1} \sum_{k=1}^{\infty} \frac{z^n}{n k^{n-1}} \\
&= (\log(2\pi) - 1) \frac{z}{2} - (1 + \gamma) \frac{z^2}{2} + \sum_{n=3}^{\infty} (-1)^{n-1} \frac{z^n}{n} \sum_{k=1}^{\infty} \frac{1}{k^{n-1}} \\
&= (\log(2\pi) - 1) \frac{z}{2} - (1 + \gamma) \frac{z^2}{2} + \sum_{n=3}^{\infty} (-1)^n \frac{z^n}{n} \zeta(n-1),
\end{aligned}$$

as required. ■

Proof of 1.14. Let us prove this result first:

Lemma 4.11.

$$f_{\text{CUE}}(s/2) = \exp \left[\frac{s^2}{4}(\gamma + 1) + \sum_{k=3}^{\infty} (-s)^k \frac{2^{k-1} - 1}{2^{k-1}} \frac{\zeta(k-1)}{k} \right], \text{ for } |s| < 1.$$

Proof. By the definition of f_{CUE} and of the cumulants, together with Proposition 3.3, we have

$$\begin{aligned}
f_{\text{CUE}}(s/2) &= \\
&= \lim_{N \rightarrow \infty} \frac{M_N(s)}{N^{s^2/4}} \\
&= \lim_{N \rightarrow \infty} \frac{1}{N^{s^2/4}} \exp \left(\sum_{j=1}^{\infty} \frac{c_j(N) s^j}{j!} \right) \\
&= \lim_{N \rightarrow \infty} \frac{1}{N^{s^2/4}} \exp \left\{ c_2(N) \frac{s^2}{2} + \sum_{k=3}^{\infty} \left[\frac{2^{k-1} - 1}{2^{k-1}} (-1)^k \right. \right. \\
&\quad \left. \left(\Gamma(k) \zeta(k-1) - \frac{(k-3)!}{N^{k-2}} \right) + O \left(\frac{1}{N^{k-1}} \right) \right] \frac{s^k}{k!} \right\} = \\
&= \lim_{N \rightarrow \infty} \frac{1}{N^{s^2/4}} \exp \left\{ \left[\frac{1}{2} \log N + \frac{1}{2}(\gamma + 1) + O(N^{-2}) \right] \frac{s^2}{2} + \right. \\
&\quad \left. \sum_{k=3}^{\infty} \left[\frac{2^{k-1} - 1}{2^{k-1}} (-1)^k \frac{s^k}{k} \zeta(k-1) - \frac{s^k}{N^{k-2} k (k-1)(k-2)} + O(N^{1-k}) \frac{s^k}{k!} \right] \right\}.
\end{aligned}$$

Now note that the factors with $O(N^{-2})$ and $O(N^{1-k})$ coefficients tend to 0 in the limit $N \rightarrow \infty$. The first term of the exponential also becomes $\frac{s^2}{4}(\gamma + 1)$. It is clear that $S = \sum_{k=3}^{\infty} \frac{s^k}{N^{k-2} k (k-1)(k-2)}$ converges, since $\left| \frac{s^k}{N^{k-2} k (k-1)(k-2)} \right| \leq \frac{1}{N k!}$ ($|s| < 1$). The triangle inequality gives $|S| \leq \frac{e}{N} \rightarrow 0$ as $N \rightarrow \infty$. So $S \rightarrow 0$ and we are left with

$$f_{\text{CUE}}(s/2) = \exp \left(\frac{s^2}{4}(\gamma + 1) + \sum_{k=3}^{\infty} (-s)^k \frac{2^{k-1} - 1}{2^{k-1}} \frac{\zeta(k-1)}{k} \right),$$

as required. ■

We want to prove 1.14 by using Lemma 4.10 in the following way:

$$\begin{aligned} \log \left(\frac{(G(1 + s/2))^2}{G(1 + s)} \right) &= 2 \log G(1 + s/2) - \log G(1 + s) \\ &= 2(\log(2\pi) - 1) \frac{s}{4} - 2(1 + \gamma) \frac{s^2}{8} + 2 \sum_{n=3}^{\infty} (-1)^{n-1} \zeta(n-1) \frac{s^n}{n 2^n} \\ &\quad - (\log(2\pi) - 1) \frac{s}{2} + (1 + \gamma) \frac{s}{2} - \sum_{n=3}^{\infty} (-1)^{n-1} \zeta(n-1) \frac{s^n}{n} = \\ &= (1 + \gamma) \frac{s^2}{4} + \sum_{n=3}^{\infty} \frac{2^{n-1} - 1}{2^{n-1}} \zeta(n-1) \frac{(-s)^n}{n}. \end{aligned}$$

Combining this with Lemma 4.11 yields $\log f_{\text{CUE}}(s/2) = \log \left(\frac{(G(1+s/2))^2}{G(1+s)} \right)$, from which the conclusion follows. ■

Remark 4.12. The above equality holds for $|s| < 1$, but we can extend it by analytic continuation to the rest of the complex plane (G is an entire function). Moreover, $f_{\text{CUE}}(s/2)$ is a meromorphic function of order 2 with a pole of order $(2k - 1)$ at each odd negative integer.

Remark 4.13. We can compute the values of f_{CUE} for positive integers n by using the recurrence relations. It is easy to show that $G(n) = \prod_{k=1}^{N-1} \Gamma(k)$ for $n \geq 2$, so

$$f_{\text{CUE}}(n) = \frac{G(1 + n)^2}{G(1 + 2n)} = \frac{\prod_{k=1}^n \Gamma(k)^2}{\prod_{m=1}^{2n} \Gamma(m)} = \frac{\prod_{k=1}^n \Gamma(k)}{\prod_{m=n+1}^{2n} \Gamma(m)} = \prod_{k=0}^{n-1} \frac{k!}{(k + n)!},$$

for $n \geq 1$. It has also been shown that the same result holds for a wider class of Hermitian random matrices, which contains the GUE [22].

The counterpart to 1.14 can be proved similarly:

Proposition 4.14. $\lim_{N \rightarrow \infty} L_N(s) N^{s^2/4} = G(1 - s/2) G(1 + s/2)$, for $|s/2| < 1$.

Proof. As before, we need a preliminary, very similar, result:

Lemma 4.15.

$$g_{\text{CUE}}(s/2) := \lim_{N \rightarrow \infty} L_N(s) N^{s^2/4} = \exp \left(-(\gamma + 1) \frac{s^2}{4} - \sum_{k=2}^{\infty} \frac{\zeta(2k-1) s^{2k}}{k 2^{2k}} \right).$$

Proof. Recall from 3.9 that

$$\begin{aligned} d_2(N) &= c_2(N) \\ d_{2n}(N) &= \frac{(-1)^{n+1}}{2^{2n-1}} \left(\Gamma(2n) \zeta(2n-1) - \frac{(2n-3)!}{N^{2n-2}} \right) + O(N^{1-2n}), \forall N \geq 2 \end{aligned}$$

and the rest of the cumulants are 0. So

$$\begin{aligned}
g_{\text{CUE}}(s/2) &= \lim_{N \rightarrow \infty} N^{s^2/4} \exp \left(\sum_{k=1}^{\infty} \frac{d_k(N)(is)^k}{k!} \right) \\
&= \lim_{N \rightarrow \infty} N^{s^2/4} \exp \left\{ - \left[\frac{1}{2} \log N + \frac{1}{2}(\gamma + 1) + O(N^{-2}) \right] \frac{s^2}{2} + \right. \\
&\quad \left. \left[\sum_{k=2}^{\infty} \frac{(-1)^{k+1}}{2^{2k-1}} \left(\Gamma(2k)\zeta(2k-1) - \frac{(2k-3)!}{N^{2k-2}} \right) + O(N^{1-2k}) \right] \frac{(-1)^k s^{2k}}{(2k)!} \right\} \\
&= \exp \left(-(\gamma + 1) \frac{s^2}{4} - \sum_{k=2}^{\infty} \frac{\zeta(2k-1)s^{2k}}{k2^{2k}} \right),
\end{aligned}$$

where we have used the same arguments as in 4.11. The difference is that we have to scale s by i , due to the way the cumulants were defined. $\blacksquare$

Now observe that

$$\begin{aligned}
\log [G(1-s/2)G(1+s/2)] &= \log G(1-s/2) + \log G(1+s/2) \\
&= -(\log(2\pi) - 1) \frac{s}{4} - (1 + \gamma) \frac{s^2}{8} + \sum_{n=3}^{\infty} (-1)^{n-1} \zeta(n-1) \frac{(-s)^n}{n2^n} \\
&\quad + (\log(2\pi) - 1) \frac{s}{4} - (1 + \gamma) \frac{s^2}{8} + \sum_{n=3}^{\infty} (-1)^{n-1} \zeta(n-1) \frac{s^n}{n2^n} \\
&= -(1 + \gamma) \frac{s^2}{4} + 2 \sum_{n=2}^{\infty} (-1)^{2n-1} \zeta(2n-1) \frac{(-s)^{2n}}{2n2^{2n}} \\
&= -(1 + \gamma) \frac{s^2}{4} - \sum_{n=2}^{\infty} \frac{\zeta(2n-1)s^{2n}}{n2^{2n}},
\end{aligned}$$

for $|s/2| < 1$. So $\log [g_{\text{CUE}}(s/2)] = \log [G(1-s/2)G(1+s/2)]$ and the conclusion follows. We can still extend the function by analytic continuation to the whole complex plane, since $g_{\text{CUE}}(s/2)$ has zeros of order n at $s = \pm 2n$, for $n \geq 1$. $\blacksquare$

4.3. COE and CSE. In the last part of this section, we will focus on deriving $f_{\text{COE}}, f_{\text{CSE}}$ and their counterparts, just like in the CUE setting. The computations are very similar to the ones in the proofs of 4.10, 4.11 and 4.15, so we shall only detail the essential steps when computing them.

In all results, the functions can be extended meromorphically (or analytically if there are no poles) to the whole complex plane. The series expansions of $G(1+z)$ and $\log \Gamma(1+z)$ only make sense when $|z| < 1$, and we will adjust the range according to the coefficients of s .

All of the subsequent results can be checked by writing $M_N^\alpha(s)$ and $L_N^\alpha(s)$ as polynomials in N , but these quantities stand out due to their significance in deriving similar central limit theorems and conjectures. Inspired by the first order asymptotics of $c_2^\alpha(N)$, we can prove, using the same techniques as in Sections 2 and 3, that a generalized version of Theorem 1.11 holds:

Theorem 4.16. *Let $R \in \mathbb{R}^2$ be a rectangle and let $Z(\alpha, U, \theta)$ be the characteristic polynomial of a $N \times N$ matrix M with respect to each ensemble, for $\alpha \in \{1, 2, 4\}$.*

Then

$$\lim_{N \rightarrow \infty} \text{meas} \left\{ M \in U_\alpha(N) : \frac{\log Z(\alpha, U, \theta)}{\sqrt{1/\alpha \log N}} \in R \right\} = \frac{1}{2\pi} \iint_R e^{-(x^2+y^2)/2} dx dy.$$

Let us derive the COE results:

Proposition 4.17. For $|s| < 1/2$,

$$f_{\text{COE}}(s) = \lim_{N \rightarrow \infty} \frac{M_N^1(s)}{N^{s^2/2}} = \frac{G(1+s)\Gamma(1+s)\sqrt{\Gamma(1+s)}}{\Gamma(1+s/2)\sqrt{G(1+2s)\Gamma(1+2s)}}$$

Proof. As in 4.11 and using the asymptotic formula for $c_2^1(\infty)$,

$$f_{\text{COE}}(s) = \exp \left\{ \left(1 + \gamma - \frac{3}{4}\zeta(2) \right) \frac{s^2}{2} + \sum_{n=3}^{\infty} (-1)^n (2^{n-1}\zeta(n-1) - \zeta(n-1) - 2^{n-1}\zeta(n) + \frac{3}{2}\zeta(n) - \frac{1}{2^n}\zeta(n)) \frac{s^n}{n} \right\}$$

Furthermore, from A.9,

$$\begin{aligned} \log \text{RHS} &= \log G(1+s) + \frac{3}{2} \log \Gamma(1+s) - \log \Gamma(1+s/2) - \frac{1}{2} \log G(1+2s) \\ &\quad - \frac{1}{2} \log \Gamma(1+2s) \\ &= (\log(2\pi) - 1) (s/2 - s/2) - (1 + \gamma) \left(\frac{s^2}{2} - \frac{1}{2} \frac{(2s)^2}{2} \right) \\ &\quad + \sum_{n=3}^{\infty} (-1)^{n-1} \zeta(n-1) (1 - 2^{n-1}) \frac{s^n}{n} + \left(1 + \gamma - \frac{3}{4}\zeta(2) \right) \frac{s^2}{2} \\ &\quad + \sum_{n=3}^{\infty} (-1)^n \zeta(n) \left(\frac{3}{2}\zeta(n) - 2^{n-1}\zeta(n) - \frac{1}{2^n}\zeta(n) \right) \frac{s^n}{n} = \\ &= \left(1 + \gamma - \frac{3}{4}\zeta(2) \right) \frac{s^2}{2} + \sum_{n=3}^{\infty} (-1)^n (2^{n-1}\zeta(n-1) - \zeta(n-1)) \frac{s^n}{n} \\ &\quad + \sum_{n=3}^{\infty} \left(-2^{n-1}\zeta(n) + \frac{3}{2}\zeta(n) - \frac{1}{2^n}\zeta(n) \right) \frac{s^n}{n}. \end{aligned}$$

Note that this only holds for $|s| < 1/2$, since A.9 is only valid for $|s| < 1$. Since the logarithms are equal, it follows that indeed

$$f_{\text{COE}}(s) = \frac{G(1+s)\Gamma(1+s)\sqrt{\Gamma(1+s)}}{\Gamma(1+s/2)\sqrt{G(1+2s)\Gamma(1+2s)}},$$

as required. ■

Remark 4.18. The function has simple poles at $s = -(2k-1)$ and a pole of order k at $s = -(2k+1)/2$, for $k \geq 1$.

Remark 4.19. f_{COE} has rational coefficients for $s = 2k$, where $k \in \mathbb{N}$. Indeed,

$$f_{\text{COE}}(2k) = \frac{G(1+2k)\Gamma(1+2k)\sqrt{\Gamma(1+2k)}}{\Gamma(1+k)\sqrt{G(1+4k)\Gamma(1+4k)}} = \prod_{j=1}^k \frac{(2j-1)!}{(2k+2j-1)!},$$

which can be proved by induction. The equality is obvious for $k = 1$. Now assume that it holds for $n \geq 1$ and we prove it for $(n + 1)$:

$$\begin{aligned}
f_{\text{COE}}(2(n + 1)) &= \Gamma(2n + 1)\Gamma(2n + 2)G(1 + 2n) [\Gamma(1 + 2n)]^{3/2} [(2n + 1)(2n + 2)]^{3/2} \\
&\quad \times \left((n + 1)\Gamma(1 + n) \left(\prod_{t=4n+1}^{4n+4} \Gamma(t)t \right)^{1/2} \sqrt{G(1 + 4n)\Gamma(1 + 4n)} \right)^{-1} \\
&= \frac{(2n)!(2n + 1)! [(2n + 1)(2n + 2)]^{3/2}}{(n + 1)(4n)!(4n + 2)!(4n + 1)(4n + 3)\sqrt{(4n + 2)(4n + 4)}} \times \\
&\quad \times \prod_{j=1}^n \frac{(2j - 1)!}{(2n + 2j - 1)!} = \\
&= \frac{[(2n + 1)!]^2 (2n + 2)^{3/2}}{(2n + 2)(4n + 1)!(4n + 3)!\sqrt{2n + 2}} \prod_{j=1}^n \frac{(2j - 1)!}{(2n + 2j - 1)!} \\
&= \prod_{j=1}^{n+1} \frac{(2j - 1)!}{(2n + 2j + 1)!}.
\end{aligned}$$

Throughout the calculations, we have used the functional equation and the simple result $G(n) = \prod_{k=1}^{n-1} \Gamma(k)$. The last identity follows from splitting the product to recover the first n factors. The induction step has been proved, so the equality holds for all $k \geq 1$.

Proposition 4.20. *For $|s| < 1$,*

$$\lim_{N \rightarrow \infty} L_N^1(s) \times N^{s^2/2} = \sqrt{\frac{G(1 + s)G(1 - s)\Gamma(1 + s)\Gamma(1 - s)}{\Gamma(1 + s/2)\Gamma(1 - s/2)}}.$$

Proof. As before, we use 4.7 and the asymptotic formula for the second cumulant (the $O(N^{-1})$ term vanishes):

$$\begin{aligned}
g_{\text{COE}}(s) &= \lim_{N \rightarrow \infty} N^{s^2/2} L_N(s) = \\
&= N^{s^2/2} \exp \left(\sum_{n=1}^{\infty} d_{2n}(\infty) \frac{(is)^n}{n!} \right) \\
&= N^{s^2/2} \exp \left\{ - \left(\log N + 1 + \gamma - \frac{3}{4}\zeta(2) \right) \frac{s^2}{2} + \sum_{n=2}^{\infty} (-1)^n \frac{s^{2n}}{(2n)!} (-1)^{3n+1} \Gamma(2n) \right. \\
&\quad \times \left. \left(\zeta(2n - 1) - \zeta(2n) + \frac{1}{2^{2n}} \zeta(2n) \right) \frac{s^{2n}}{(2n)!} \right\} = \\
&= \exp \left\{ - \left(1 + \gamma - \frac{3}{4}\zeta(2) \right) \frac{s^2}{2} - \sum_{n=2}^{\infty} \left(\zeta(2n - 1) - \zeta(2n) + \frac{1}{2^{2n}} \zeta(2n) \right) \frac{s^{2n}}{2n} \right\}.
\end{aligned}$$

Compare $\log g_{\text{COE}}(s)$ now with

$$\log \text{RHS} = \frac{1}{2} \{ \log G(1 + s) + \log G(1 - s) + \log \Gamma(1 + s) + \log \Gamma(1 - s) -$$

$$\begin{aligned}
& -\log \Gamma(1 + s/2) - \log \Gamma(1 - s/2)\} = \\
& = \frac{1}{2} \left\{ -(1 + \gamma)s^2 + \sum_{n=2}^{\infty} [(-1)^{2n-1} - 1] \zeta(2n-1) \frac{s^{2n}}{2n} \right. \\
& \quad \left. + \sum_{n=1}^{\infty} (-1)^{2n} \zeta(n) \left(\frac{s^{2n}}{2n} + \frac{(-s)^{2n}}{2n} - \frac{s^{2n}}{2n2^{2n}} - \frac{(-s)^{2n}}{2n2^{2n}} \right) \right\} = \\
& = -\left(1 + \gamma - \frac{3}{4}\zeta(2)\right) \frac{s^2}{2} - \sum_{n=2}^{\infty} \left(\zeta(2n-1) - \zeta(2n) + \frac{1}{2^{2n}}\zeta(2n) \right) \frac{s^{2n}}{2n}.
\end{aligned}$$

The function has zeros of order k at all nonzero integers. ■

We now compute the same quantities for the CSE case. The calculations are very similar, but we go through the main steps for completeness:

Proposition 4.21. For $|s| < 1$,

$$f_{\text{CSE}}(s) = \lim_{N \rightarrow \infty} \frac{M_N^4(s)}{N^{s^2/2}} = 2^{s^2/8} \frac{G(1 + s/2)\Gamma(1 + s/4)\sqrt{\Gamma(1 + s)}}{\sqrt{G(1 + s)\Gamma(1 + s/2)\Gamma(1 + s/2)}},$$

Proof. Again, use the previous results to get

$$\begin{aligned}
f_{\text{CSE}}(s) &= \lim_{N \rightarrow \infty} \frac{M_N^4(s)}{N^{s^2/8}} \\
&= \exp \left\{ \left(\frac{1}{4}(1 + \gamma) + \frac{1}{4} \log 2 + \frac{3}{16}\zeta(2) \right) \frac{s^2}{2} + \sum_{n=3}^{\infty} (-1)^n \frac{s^n}{n} \right. \\
&\quad \left. \times \left(\frac{1}{2} - \frac{1}{2^n} \right) \left(\zeta(n-1) + \zeta(n) - \frac{1}{2^n}\zeta(n) \right) \right\} = \\
&= \exp \left\{ \left(\frac{1}{4}(1 + \gamma) + \frac{1}{4} \log 2 + \frac{3}{16}\zeta(2) \right) \frac{s^2}{2} + \sum_{n=3}^{\infty} (-1)^n \frac{s^n}{n} \right. \\
&\quad \left. \times \left(\frac{1}{2}\zeta(n-1) - \frac{1}{2^n}\zeta(n-1) + \frac{1}{2}\zeta(n) - \frac{1}{2^n}\zeta(n) - \frac{1}{2^{n+1}}\zeta(n) + \frac{1}{4^n}\zeta(n) \right) \right\}.
\end{aligned}$$

On the other hand,

$$\begin{aligned}
\log \text{RHS} &= \frac{s^2}{8} \log 2 + \log G(1 + s/2) + \frac{1}{2} \log \Gamma(1 + s) + \log \Gamma(1 + s/4) - \\
&\quad - \frac{1}{2} \log G(1 + s) - \frac{3}{2} \log \Gamma(1 + s/2) = \\
&= (1 + \gamma) \frac{s^2}{8} + \sum_{n=3}^{\infty} (-1)^{n-1} \zeta(n-1) \left(\frac{s^n}{2^n n} - \frac{s^n}{2n} \right) - \\
&\quad - s\gamma + \frac{s}{4}\gamma + \frac{3}{4}s\gamma + \sum_{n=2}^{\infty} (-1)^n \zeta(n) \left(\frac{s^n}{2n} + \frac{s^n}{4^n n} - \frac{3s^n}{2^{n+1}n} \right) = \\
&= \frac{s^2}{8} \log 2 + \left(\frac{1}{4}(1 + \gamma) + \frac{3}{16}\zeta(2) \right) \frac{s^2}{2} + \sum_{n=3}^{\infty} (-1)^n \frac{s^n}{n} \times \\
&\quad \times \left(\frac{1}{2}\zeta(n-1) - \frac{1}{2^n}\zeta(n-1) + \frac{1}{2}\zeta(n) - \frac{1}{2^n}\zeta(n) - \frac{1}{2^{n+1}}\zeta(n) + \frac{1}{4^n}\zeta(n) \right).
\end{aligned}$$

So $\log \text{RHS} = \log f_{\text{CSE}}(s)$ and the equality holds for $|s| < 1$. The function has poles of order k at odd negative integers and a zero of order 1 at $-4k + 2$, for $k \geq 1$. ■

Remark 4.22. Once again, note that the rational values are those for which $s = 4k$, for positive integers k . In this case we get

$$f_{\text{CSE}}(4k) = \frac{2^k}{\left(\prod_{j=1}^{2k-1} (2j-1)!!\right) (2k-1)!!}.$$

We can use induction on $n \geq 1$ again. First, note that

$$\begin{aligned} \frac{2^{n+1}}{\left(\prod_{j=1}^{2n+1} (2j-1)!!\right) (2n+1)!!} &= \frac{2}{(2n+1)(4n-1)!!(4n+1)!!} \frac{2^n}{\left(\prod_{j=1}^{2n-1} (2j-1)!!\right) (2n-1)!!} \\ &= \frac{2}{(2n+1)(4n-1)!!(4n+1)!!} f_{\text{CSE}}(4n). \end{aligned}$$

Then observe that, for $n \geq 1$, by elementary properties of functions,

$$\begin{aligned} f_{\text{CSE}}(4n+4) &= 2^{4n+2} \Gamma(2n+2) \Gamma(2n+1) (n+1) \left(\prod_{k=4n+1}^{4n+4} k \right)^{1/2} \times \\ &\quad \times \left\{ \left(\prod_{k=4n+1}^{4n+4} \Gamma(k) \right)^{1/2} [(2n+2)(2n+1)]^{3/2} \right\}^{-1} f_{\text{CSE}}(4n) = \\ &= \frac{2^{4n+2} (2n+1)! (2n)! (n+1) \sqrt{(4n+4)(4n+2)}}{(4n)! (4n+2)! [(2n+2)(2n+1)]^{3/2}} f_{\text{CSE}}(4n) \\ &= \frac{2^{4n+2} ((2n)!)^2}{(4n)! (4n+2)!} f_{\text{CSE}}(4n) = \frac{2}{(2n+1)(4n-1)!!(4n+1)!!} f_{\text{CSE}}(4n) \\ &= \frac{2^{n+1}}{\left(\prod_{j=1}^{2n+1} (2j-1)!!\right) (2n+1)!!}, \end{aligned}$$

so the induction step has been proved.

The last result of this section is

Proposition 4.23. For $|s| < 2$,

$$g_{\text{CSE}}(s) = \lim_{N \rightarrow \infty} L_N^4(s) \times N^{s^2/8} = 2^{-s^2/8} \sqrt{\frac{G(1+s/2)G(1-s/2)\Gamma(1+s/4)\Gamma(1-s/4)}{\Gamma(1+s/2)\Gamma(1-s/2)}}.$$

Proof. This is done in exactly the same way. By the asymptotic formula and noting that the expansion is with i s instead of s ,

$$\begin{aligned} g_{\text{CSE}}(s) &= \exp \left\{ - \left(\frac{1}{4}(1+\gamma) + \frac{1}{4} \log 2 + \frac{3}{16} \zeta(2) \right) \frac{s^2}{2} - \right. \\ &\quad \left. - \sum_{n=2}^{\infty} \left(\frac{1}{2^{2n}} \zeta(2n-1) + \frac{1}{2^{2n}} \zeta(2n) - \frac{1}{4^{2n}} \zeta(2n) \right) \frac{s^{2n}}{2n} \right\} \end{aligned}$$

Expanding the logarithm of the right hand side gives the same expression as in the exponential. ■

Remark 4.24. The function has zeros of order k at $\pm(4k - 2)$ and $\pm 4k$, for $k \geq 1$, so it can be extended analytically.

5. NUMERICAL DATA AND ASYMPTOTIC ESTIMATIONS OF MOMENTS

In this section we largely follow the structure of [1], with a few additions, and present the main numerical estimates that have inspired the previous results and we provide further context for other conjectures. In this sense, it can be thought of as a more detailed continuation of the Introduction.

The first important observation (recalled in the Introduction) is that we can relate the matrix size N with the height T used for counting the zeros on the critical line, so that Theorems 1.10 and 1.11 have a clear correspondence. This is given, on one hand, by the mean density $2\pi/N$ of the eigenphases (see 1.1) and by the mean spacing between the Riemann zeros $2\pi/\log(T/2\pi)$, on the other (as claimed in the beginning of the Introduction). It is, thus, only natural to make the identification

$$N = \log \frac{T}{2\pi},$$

where T is taken so that N is an integer. In fact, this is the only property that also appears in Montgomery's conjecture and its generalizations.

Montgomery's conjecture, however, concerns only correlations on the scale of the mean density of the Riemann zeros. The long-range correlations can be written as a sum of two contributions, in a similar way to Conjecture 1.12. This is also true for the second moment of $\text{Im} \log \zeta(1/2 + it)$, where these split up into a random matrix limit and a convergent series over the primes. This is known as Goldston's formula [23]:

$$\begin{aligned} \frac{1}{T} \int_0^T (\text{Im} \log \zeta(1/2 + it))^2 dt &= \frac{1}{2} \log \log \frac{T}{2\pi} + \frac{1}{2}(\gamma + 1) + \\ &+ \sum_{n=2}^{\infty} \sum_{p \text{ prime}} \frac{1-n}{n^2 p^n} + o(1). \end{aligned}$$

We also expect a connection between the moments of $\zeta(1/2 + it)$ averaged over t and the moments of $|Z(U, \theta)|$ averaged over the CUE. Since the zeta function can be expressed as a convergent product over the primes, we expect the moments to be evaluated in terms of primes as well. This is supported by a conjecture of Heath-Brown [24], similar to 1.12, which asserts that, for

$$I_k(T, \sigma) = \int_1^T |\zeta(\sigma + it)|^{2k} dt$$

and multiplicative functions $d_k(n)$, defined for $k \geq 0$ by

$$\zeta^k(s) = \prod_{p \text{ prime}} \left(1 - \frac{1}{p^s}\right)^{-k} = \sum_{n=1}^{\infty} d_k(n) n^{-s}, \text{ we have}$$

$$I_k(T, \sigma) \sim \sum_{n=1}^{\infty} \frac{d_k(n)^s}{n^{2\sigma}}, \text{ as } t \rightarrow \infty.$$

There is a nice way to exhibit the closeness of the numerical values from the number theoretic and the CUE settings. The convergence guaranteed by Selberg's Theorem is slow, as shown by A. Odlyzko's computations for the value distributions of $\text{Re} \log \zeta(1/2 + it)$ and $\text{Im} \log \zeta(1/2 + it)$ near the 10^{20} th zero [7]. These correspond to a matrix size $N = 42$ and exhibit striking deviations from the Gaussian limiting distribution, but significantly smaller ones compared to the real and imaginary parts

of $\log Z / \sqrt{1/2 \log N}$ (the CUE moments). It is worth remarking that, even though the value distribution of $\text{Im } \log Z$ is symmetric about 0, the distribution of $\text{Re } \log Z$ is not and shows larger deviations.

It is instructive to test Conjectures 1.12 and 1.15 by looking at the moments of the real and imaginary parts of $\log \zeta(1/2 + it)$, under the identification $N = \log \frac{T}{2\pi}$. In the real case, we expect to have

$$\frac{1}{T} \int_0^T [\text{Re } \log \zeta(1/2 + it)]^k dt \sim \frac{d^k}{ds^k} [M_N(s)a(s/2)]_{s=0},$$

according to numerical data presented in [1].

The other moment is related to the generating function of $\text{Im } \log Z$ in a similar way (we still need to scale by $-i$, as everything was defined in terms of $L_N(-is)$):

$$\frac{1}{T} \int_0^T [\text{Im } \log \zeta(1/2 + it)]^k dt \sim (-i)^k \frac{d^k}{ds^k} [L_N(s)b(s/2)]_{s=0}, \text{ where}$$

$$b(\lambda) = \prod_{p \text{ prime}} \left[\left(1 - \frac{1}{p}\right)^{\lambda^2} \sum_{n=0}^{\infty} \frac{\Gamma(1+\lambda)\Gamma(1-\lambda)}{\Gamma(1+\lambda-n)\Gamma(1-\lambda-n)(n!)^2} p^{-n} \right].$$

Observe that $b(\lambda)$ is very similar to $a(\lambda)$. Both are derived in the same way and more intuition is provided by [14]. The authors made the conjectures for the sixth and eight moments based on an approximate functional equation for $\zeta(s)^k$, which involves the sum of two Dirichlet polynomials.

Inspired by Proposition 4.14, we can formulate the counterpart to Conjecture 1.12:

Conjecture 5.1.

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \left(\frac{\zeta(1/2 + it)}{\zeta(1/2 - it)} \right)^\lambda dt = G(1 - \lambda)G(1 + \lambda)b(\lambda),$$

for all suitable λ .

We would now like to find similarities between the characteristic polynomial and the distribution of $|\zeta(1/2 + it)|$, under the identification $N = \log \frac{T}{2\pi}$. This is done by looking at the density of $|Z|$,

$$\tau_N(t) = [\delta(t - |Z|)]_{U(N)}.$$

This is given by

$$\begin{aligned} \tau_N(t) &= \frac{1}{t} [\delta(\log t - \log |Z|)]_{U(N)} \\ &= \frac{1}{2\pi t} \left[\int_{-\infty}^{\infty} e^{-is(\log t - \log |Z|)} ds \right]_{U(N)} \\ &= \frac{1}{2\pi t} \int_{-\infty}^{\infty} e^{-is \log t} M_N(is) ds, \end{aligned}$$

upon making the transformation $t \rightarrow \log t$ (we assume t is positive). We can then expand $M_N(is)$ and retain only the second cumulant, just like we did in 3.5 and 3.6,

and the corresponding calculations are the same as in [1] :

$$\begin{aligned}\tau_N(t) &= \frac{1}{2\pi t} \exp \left(-is \log t - c_2(N) \frac{s^2}{2} - ic_3(N) \frac{s^3}{3!} + \dots \right) ds \\ &\sim \frac{1}{2\pi t \sqrt{c_2(N)}} \int_{-\infty}^{\infty} \exp \left(-\frac{is \log t}{\sqrt{c_2(N)}} - \frac{s^2}{2} \right) \\ &= \frac{1}{t \sqrt{2\pi c_2(N)}} \exp \left(-\frac{\log^2 t}{2c_2(N)} \right),\end{aligned}$$

which holds for t fixed and $N \rightarrow \infty$.

Recall that $M_N(s) = \prod_{k=1}^N \frac{\Gamma(k)\Gamma(s+k)}{(\Gamma(k+s/2))^2}$, for $\operatorname{Re}(s) > -1$. Note that this function has poles at negative integers $s = -n$, for $n \geq 1$, and can be extended meromorphically to $\mathbb{C}$. We can relate this quantity to the following result in an interesting way, as we shall see in 5.3.

Proposition 5.2.

$$\lim_{t \rightarrow 0} \tau_N(t) = \frac{1}{\Gamma(N)} \prod_{k=1}^N \left(\frac{\Gamma(k)}{\Gamma(k-1/2)} \right)^2.$$

Proof. The derivation is complicated and a proof of a very similar integral is outlined after Theorem 3.9 of [25]. Let $f(s) := \frac{1}{2\pi t} e^{-is \log t} M_N(is)$, where we use the meromorphic extension of M_N . From the functional equation, f has poles of order n at $s = in$, for $1 \leq n \leq N$.

The main idea is to integrate f over a rectangle with vertices $-R, R, R+i+\epsilon i, -R+i+\epsilon i$, for $0 < \epsilon < 1$ and $R \rightarrow \infty$. The only pole is at $s = i$ and we can use Cauchy's Residue Theorem:

$$\begin{aligned}\int_R f(z) dz &= 2\pi i \operatorname{Res}(f; i) \\ &= \frac{i}{t} \lim_{s \rightarrow i} (s-i) e^{-is \log t} \prod_{k=1}^N \frac{\Gamma(k)\Gamma(is+k)}{(\Gamma(k+is/2))^2} \\ &= \prod_{k=2}^N \frac{\Gamma(k)\Gamma(k-1)}{(\Gamma(k-1/2))^2} \lim_{s \rightarrow i} (is+1) \frac{\Gamma(1+is)}{(\Gamma(1/2))^2} \\ &= \frac{1}{\Gamma(N)} \prod_{k=1}^N \left(\frac{\Gamma(k)}{\Gamma(k-1/2)} \right)^2,\end{aligned}$$

since $(is+1)\Gamma(is+1) = \Gamma(is+2)$.

One of the side integrals is

$$I_1 := \frac{1}{2\pi t} \int_{-R}^{-R+(1+\epsilon)i} e^{-is \log t} M_N(is) ds,$$

from which we get

$$|I_1| \leq \frac{1}{2\pi t} \int_{-R}^{-R+(1+\epsilon)i} e^{is \log t} |M_N(is)| ds.$$

An equivalent expression for $M_N(s)$ is $\frac{G^2(1+s/2)G(N+1)(N+1+s)}{(1+s)G^2(N+1+s/2)}$ [25]. By the asymptotics of the G-function, we expect $|M_N(is)|$ to be bounded between the integration limits, in which case we can use the Estimation Theorem to see that the integral vanishes as $R \rightarrow \infty$, $t \rightarrow 0$. The other integral has a similar expression and also vanishes [25]. We are therefore left with

$$\int_R f(s) ds = \int_{-\infty}^{\infty} f(s) ds + \int_{-\infty+(1+\epsilon)i}^{\infty+(1+\epsilon)i} f(s) ds.$$

However, we have not been able to make progress on the subsequent calculations. ■

We can now obtain the leading-order asymptotics like in the proof of Theorem 1.14. Specifically,

Proposition 5.3.

$$\lim_{t \rightarrow 0} \frac{\tau_N(t)}{N^{1/4}} = G^2(1/2).$$

Proof. Plugging $\lambda = 1$ in 1.14 gives

$$f_{\text{CUE}}(1/2) = \lim_{N \rightarrow \infty} \frac{M_N(1)}{N^{1/4}} = (G(3/2))^2 / G(2) = (\Gamma(1/2))^2 (G(1/2))^2. \text{ However,}$$

$$\begin{aligned} M_N(1) &= \left(\frac{\Gamma(1/2)}{\Gamma(N+1/2)} \right)^2 \Gamma(N+1)\Gamma(N) \frac{1}{\Gamma(N)} \prod_{k=1}^N \left(\frac{\Gamma(k)}{\Gamma(k-1/2)} \right)^2 \\ &= \frac{\Gamma(N+1)\Gamma(N)}{(\Gamma(N+1/2))^2} (\Gamma(1/2))^2 \tau_N(t). \end{aligned}$$

So $\lim_{N \rightarrow \infty} \frac{M_N(1)}{N^{1/4}} = (\Gamma(1/2))^2 \lim_{N \rightarrow \infty} \frac{\tau_N(t)}{N^{1/4}}$ (the fraction clearly tends to 1), from which the conclusion follows. ■

We can also derive an explicit expression for $G(1/2)$:

Proposition 5.4.

$$G(1/2) = \exp \left(\frac{1}{24} \log 2 + \frac{3}{2} \zeta'(-1) - \frac{1}{4} \log \pi \right).$$

Proof. The proof is outlined in the appendix of [26]. First, the authors claim that

$$G(1/2) = A^{-3/2} \pi^{-1/4} e^{1/8} 2^{1/24},$$

where A is the Glaisher constant, defined via

$$\log A = \left\{ \lim_{N \rightarrow \infty} \log \left(\prod_{n=1}^N n^n \right) - \left(\frac{N^2}{2} + \frac{N}{2} + \frac{1}{12} \right) \log N + \frac{N^2}{4} \right\}.$$

It is also proved in pages 332-3 of [27], by using the Euler-Maclaurin formula, that

$$\zeta(s) = \lim_{N \rightarrow \infty} \left\{ \sum_{n=1}^N \frac{1}{n^s} - \frac{N^{1-s}}{1-s} - \frac{N^{-s}}{2} + \frac{sN^{-s-1}}{12} \right\}, \text{ for } \text{Re}(s) > -3.$$

Differentiating under the limit at $s = -1$ gives [26]

$$\zeta'(-1) = \lim_{N \rightarrow \infty} \left\{ - \sum_{n=1}^N n \log n + \log N \left(\frac{N^2}{2} + \frac{N}{2} + \frac{1}{12} \right) - \frac{N^2}{4} + \frac{1}{12} \right\}$$

and it follows immediately that $\log A = -\zeta'(-1) + \frac{1}{12}$. So

$$\begin{aligned} \log G(1/2) &= -\frac{3}{2} \log A - \frac{1}{4} \log \pi + \frac{1}{8} + \frac{1}{24} \log 2 \\ &= \frac{3}{2} \zeta'(-1) - \frac{1}{4} \log \pi + \frac{1}{24} \log 2, \end{aligned}$$

which concludes the proof. ■

Remark 5.5. The previous results yield $\tau_N(t) \sim \exp \left(\frac{1}{12} \log 2 + 3\zeta'(1) - \frac{1}{2} \log \pi \right) N^{1/4}$ as $t \rightarrow 0$.

As in the previous conjectures about the moments of the real and imaginary parts of $\zeta(1/2 + it)$, we expect ([1]) the prime contribution of $|\zeta(1/2 + it)|$ to be given by

$$\tilde{\tau}_N(t) = \frac{1}{2\pi t} \int_{-\infty}^{\infty} e^{-is \log t} a(is/2) M_N(is) ds.$$

In the limit $t \rightarrow 0$, as in 5.2, the contribution from the pole of $M_N(is)$ at $s = i$ is dominant and we obtain the same limiting value, multiplied by $a(-1/2)$ (as this is not involved in the residue calculation):

$$\tilde{\tau}_N(0) = a(-1/2) \lim_{t \rightarrow 0} \tau_N(t).$$

This formula is in accordance with the computed numerical values of the zero density.

APPENDIX A. USEFUL RESULTS

Proposition A.1. For $0 < x < 2\pi$, $\sum_{k=1}^{\infty} \frac{\sin kx}{k} = \frac{\pi - x}{2}$.

Proof. Clearly $\sin kx = (e^{ikx} - e^{-ikx})/2i$, so we would like to show that $P(z) = \sum_{k=1}^{\infty} \frac{z^k}{k}$ converges for $z = e^{\pm ix}$. Take $z = re^{\pm ix}$, for some $r \in (0, 1)$. The series converges for $|z| < 1$, so we can use Abel's Theorem for complex power series. Taking $r \uparrow 1$ yields $P(z) \uparrow P(e^{\pm ix})$, and now we can use the expansion $\log(1 - z) = -\sum_{k=1}^{\infty} \frac{z^k}{k}$ (valid in the range of z). We get

$$\begin{aligned} \sum_{k=1}^{\infty} \frac{\sin kx}{k} &= \frac{\log(1 - e^{-ix}) - \log(1 - e^{ix})}{2i} \\ &= \frac{1}{2i} \log \frac{1 - e^{-ix}}{1 - e^{ix}} = \frac{1}{2i} \log e^{i(\pi - x)} = \frac{\pi - x}{2}, \end{aligned}$$

as required. ■

Theorem A.2 (Euler-Maclaurin Formula). For a C^{2m} function f defined on $[a, b]$, the following equality holds:

$$\sum_{p=a}^b f(p) = \frac{1}{2} (f(a) + f(b)) + \int_a^b f(x) dx + \sum_{k=1}^m \frac{B_{2k}}{(2k)!} (f^{(2k-1)}(b) - f^{(2k-1)}(a)) + R_{2m},$$

where $R_{2m} = -\frac{1}{(2m)!} \int_a^b b_{2m}(x - \lfloor x \rfloor) f^{(2m)}(x) dx$, $b_n(x)$ are the Bernoulli polynomials and B_n are the Bernoulli numbers, obtained via the expansion $\frac{x}{e^x - 1} = \sum_{n=0}^{\infty} \frac{B_n x^n}{n!}$ ($B_n = 0$ for n odd).

Proposition A.3. $H_n \sim \log n + \gamma + \frac{1}{2n}$, for large n .

Proof. Apply Theorem A.2 with $f(x) = 1/x$, $a = n$ and $b = n + p$, for p large. We can prove by induction that

$$f^{(k)}(x) = \frac{(-1)^k k!}{x^{k+1}},$$

$\forall x \neq 0$ and $k \geq 0$. Now note that

$$\begin{aligned} H_{n-1} &= \sum_{k=1}^{n+p} \frac{1}{k} - \sum_{k=n}^{n+p} \frac{1}{k} = \sum_{k=1}^{n+p} \frac{1}{k} - \log(n+p) + \log n - \frac{1}{2} \left(\frac{1}{n} + \frac{1}{n+p} \right) \\ &\quad - \sum_{r=1}^m \frac{B_{2r}}{(2r)!} \left(\frac{(2r-1)!}{(n+p)^{2r}} - \frac{(2r-1)!}{n^{2r}} \right) - R_{2m}. \end{aligned}$$

Since p was chosen arbitrarily, letting $p \rightarrow \infty$ and using the well-known result $\lim_{n \rightarrow \infty} (H_n - \log n) = \gamma$ yields

$$H_{n-1} = \gamma + \log n - \frac{1}{2n} + \sum_{r=1}^m \frac{B_{2r}}{2r n^{2r}} - R_{2m},$$

with $R_{2m} = -\int_n^{\infty} \frac{B_{2m}(x - \lfloor x \rfloor)}{x^{2m+1}} dx$, $\forall m > 0$. The remaining terms vanish when $m, n \rightarrow \infty$ (we can take any m because f is infinitely differentiable) and we derive the desired asymptotic expression for H_n immediately. ■

Remark A.4. The remaining terms are equal to ([1]) $\sum_{k=2}^{\infty} \frac{A_k}{n(n+1)\dots(n+k-1)}$,

where $A_k = \frac{1}{k} \int_0^1 x(1-x)(2-x)\dots(k-1-x) dx$.

Proposition A.5.

$$\lim_{n \rightarrow \infty} n \sum_{k=n+1}^{\infty} \frac{1}{k^2} = 1.$$

Proof. Denote the limit by l . Then

$$l = \lim_{n \rightarrow \infty} n \left(\sum_{k=1}^{\infty} \frac{1}{k^2} - \sum_{k=1}^n \frac{1}{k^2} \right) \stackrel{\text{not}}{=} \lim_{n \rightarrow \infty} n \left(\frac{\pi^2}{6} - S_n \right),$$

where $S_n - S_{n-1} = \frac{1}{n^2}$, $\forall n \geq 1$. Now we can apply the following result, due to [Stolz](#), for the $\frac{0}{0}$ case:

Lemma A.6. *Let $(a_n)_{n \geq 1}$ and $(b_n)_{n \geq 1}$ be two real sequences that converge to 0, with $(b_n)_{n \geq 1}$ strictly decreasing. If the limit $\lim_{n \rightarrow \infty} \frac{a_{n+1} - a_n}{b_{n+1} - b_n}$ exists and equals l , then $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = l$ as well.*

The corresponding sequences are $a_n = \frac{\pi^2}{6} - S_n$, $b_n = \frac{1}{n}$. Then

$$\lim_{n \rightarrow \infty} \frac{\left(\frac{\pi^2}{6} - S_{n+1} \right) - \left(\frac{\pi^2}{6} - S_n \right)}{\frac{1}{n+1} - \frac{1}{n}} = \lim_{n \rightarrow \infty} \frac{\frac{1}{(n+1)^2}}{\frac{1}{n(n+1)}} = 1.$$

Hence the Stolz Lemma ensures the original limit is 1, as required. ■

Proposition A.7. *For $\operatorname{Re}(z) > 0$, $\psi^{(1)}(z+1) = \psi^{(1)}(z) - \frac{1}{z^2}$.*

Proof. We only need the definition of the polygamma function and the functional equation $\Gamma(z+1) = z\Gamma(z)$. It is easy to see that $\Gamma'(z+1) = \Gamma'(z) + z\Gamma''(z)$ and $\Gamma''(z+1) = 2\Gamma'(z) + z\Gamma''(z)$. Since $\psi^{(1)}(z) = \frac{\Gamma''(z)}{\Gamma(z)} - \left(\frac{\Gamma'(z)}{\Gamma(z)} \right)^2$, we have

$$\begin{aligned} \psi^{(1)}(z+1) &= \frac{\Gamma''(z+1)}{\Gamma(z+1)} - \left(\frac{\Gamma'(z+1)}{\Gamma(z+1)} \right)^2 \\ &= \frac{2\Gamma'(z) + z\Gamma''(z)}{z\Gamma(z)} - \left(\frac{\Gamma'(z) + z\Gamma'(z)}{z\Gamma(z)} \right)^2 \\ &= 2\frac{\Gamma'(z)}{z\Gamma(z)} + \frac{\Gamma''(z)}{\Gamma(z)} - \frac{1}{z^2} - \left(\frac{\Gamma'(z)}{\Gamma(z)} \right)^2 - 2\frac{\Gamma'(z)}{z\Gamma(z)} \\ &= \frac{\Gamma''(z)}{\Gamma(z)} - \left(\frac{\Gamma'(z)}{\Gamma(z)} \right)^2 - \frac{1}{z^2} = \psi^{(1)}(z) - \frac{1}{z^2}, \end{aligned}$$

as required. ■

Remark A.8. If we differentiate [A.7](#) further and use induction, we get

$$\psi^{(n)}(z+1) = \psi^{(n)}(z) + \frac{(-1)^n n!}{z^{n+1}}, \quad \forall n \in \mathbb{N}.$$

Lemma A.9 (Series representation). For $n \in \mathbb{N}^*$,

$$\psi^{(n)}(z) = (-1)^{n+1} n! \sum_{k=0}^{\infty} \frac{1}{(z+k)^{n+1}}.$$

Proof. The integral equation gives

$$\begin{aligned} \psi^{(n)}(z) &= (-1)^{n+1} \int_0^{\infty} \frac{t^n e^{-zt}}{1 - e^{-t}} dt \\ &= (-1)^{n+1} \int_0^{\infty} t^n e^{-zt} \sum_{k=0}^{\infty} e^{-kt} dt \\ &= (-1)^{n+1} \sum_{k=0}^{\infty} \int_0^{\infty} t^n e^{-(z+k)t} dt \\ &= (-1)^{n+1} n! \sum_{k=0}^{\infty} \frac{1}{(z+k)^{n+1}}, \end{aligned}$$

as required. ■

Proposition A.10. For $|z| < 1$,

$$\log \Gamma(1+z) = -z\gamma + \sum_{n=2}^{\infty} (-1)^n \zeta(n) \frac{z^n}{n}.$$

Proof. We Taylor expand about $z = 0$ and then use [A.9](#) as follows:

$$\begin{aligned} \log \Gamma(1+z) &= \log \Gamma(1+0) + z \frac{d}{dz} \log \Gamma(1+z) \Big|_{z=0} + \frac{z^2}{2!} \frac{d^2}{dz^2} \log \Gamma(1+z) \Big|_{z=0} + \dots \\ &= z \psi^{(0)}(1+z) \Big|_{z=0} + \sum_{n=2}^{\infty} \frac{z^n}{n!} \psi^{(n-1)}(1+z) \Big|_{z=0} \\ &= z \psi^{(0)}(1+z) \Big|_{z=0} + \sum_{n=2}^{\infty} \frac{z^n}{n!} (-1)^n (n-1)! \sum_{k=0}^{\infty} \frac{1}{k^n} \\ &= z \psi^{(0)}(1+z) \Big|_{z=0} + \sum_{n=2}^{\infty} (-1)^n \zeta(n) \frac{z^n}{n} \end{aligned}$$

Now we can use the [expansion](#) $\psi^{(0)}(1+z) = -\gamma + \sum_{k=1}^{\infty} (-1)^{k+1} \zeta(k+1) z^k$, for $|z| < 1$.

Since the sum is evaluated at $z = 0$, it follows that

$$\log \Gamma(1+z) = -z\gamma + \sum_{n=2}^{\infty} (-1)^n \zeta(n) \frac{z^n}{n},$$

as required. ■

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